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VAR Model with Sparse Group LASSO for Multi-population Mortality Forecasting☆

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ABSTRACT

We introduce a spatial–temporally weighted vector autoregressive (SWVAR) model for modeling and forecasting mortality rates across multiple populations. First, we stack the mortality rates of the populations and build a vector autoregressive (VAR) model. Next, we apply the sparse group least absolute shrinkage and selection operator (sparse group LASSO) for fitting to avoid overparameterization. Furthermore, we integrate spatial–temporal weights, derived from age differences and geographic centroid distances, into the grouped penalty term. These approaches allow the resulting model to effectively combine information from multiple populations and reduce confounding factors associated with combined modeling. We demonstrate through a series of empirical experiments that the spatial–temporally weighted VAR model enhances estimation accuracy and exhibits superior in-sample fitting and out-of-sample forecasting performance.

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1. Introduction

Mortality modeling and forecasting have become standard tools for assessing longevity risk. Over the past decades, stochastic approaches in single populations have rapidly evolved, with prominent examples including the Lee–Carter model (Lee & Carter, 1992), the Cairns-Blake-Down model (Cairns et al., 2006), the age-period-cohort model (Renshaw & Haberman, 2006), and various extensions of the Lee–Carter model (see, e.g., Booth et al., 2006).

Modeling mortality rates across multiple populations *jointly* has received increasing attention in the literature (see, e.g., Li & Lee, 2005, Dowd et al., 2011, Hyndman et al., 2013, Boonen & Li, 2017). In these models, aggregate factors capture the common trend in mortality variations, which are considered systematic mortality developments. On the other hand, population-specific factors are used to describe the deviations of a single population from the common trend. Although factor-based models effectively address the high-dimensionality challenge of mortality data and have been successfully applied in various real-world scenarios, they still have limitations. Due to fixed age effects, these models may produce divergent mortality forecasts for different ages, and while incorporating cohort effects can improve forecast accuracy, it introduces identifiability issues. Recently, new methodologies have been developed for mortality modeling. Lin and Tsai (2022), Barigou et al. (2023), and Wang et al. (2023) use Bayesian statistical methods to improve mortality models. Machine learning and deep learning approaches have also been explored. Perla and Scognamiglio (2023) introduce a neural network fitting process, and Li et al. (2024)

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study boosting methods to enhance the classical Lee–Carter model and extend it to multiple populations. However, the issue of overfitting in the boosting techniques limits their long-term forecast reliability.

Significant advances have been made to improve traditional factor-based models in forecasting mortality rates. A notable approach involves evaluating age–period dependencies using the VAR model (e.g., Lazar & Denuit, 2009, Zhou et al., 2014, Salhi & Loisel, 2017). Subsequent work has sought to improve the VAR model through carefully designed structures or effective dimensionality reduction techniques (see Chang & Shi, 2021; Guibert et al., 2019; Li & Lu, 2017). Among them, Li and Lu (2017) propose a spatial–temporal autoregressive (STAR) model with a sparse structure that considers the dependence between historical values (temporality) and neighboring ages (which they refer to as "spatiality"). As an alternative, Guibert et al. (2019) propose a sparse VAR (SVAR) model using an elastic net approach (ENET, Zou & Hastie, 2005). However, these models lack thorough consideration of population attributes, limiting their effectiveness in multi-population mortality modeling. Li and Shi (2021) study a vector autoregressive model with exogenous variables (VARX) to focus on multi-population models. However, the global exogenous variable in their model, which is invariant across all populations, may not fully capture the complex correlation structures between populations.

Overparameterization is a primary concern in mortality modeling and forecasting, a problem that has persisted for a long time (e.g., Congdon, 1993). The number of parameters often exceeds the sample size, which is exacerbated in multi-population models. Therefore, an appropriate reduction of dimensions is needed. In Cupido et al. (2024), the authors compare a range of factor-based models, including the Li-Lee model and its extensions that incorporate GDP and spatial patterns, and select models through the Akaike Information Criterion (AIC, Akaike, 1973) and the Bayesian Information Criterion (BIC, Schwarz, 1978). Even the best model selected using BIC has a parameter-to-sample size ratio exceeding 10%, which is rather large. However, VAR models are also prone to overparameterization (see, e.g., Canova, 1999; Nicholson et al., 2020). Two econometric solutions can be employed: assigning a fixed ad hoc structure (such as in the STAR model) or using data-driven model reduction methods (such as in the SVAR model). We propose a data-driven reduction model and apply it to a multi-population setting. Our approach can reduce the number of parameters in a VAR model to less than half of those in the Li-Lee model.

The key objective of this study is to investigate methodologies for effectively addressing the persistent issue of overparameterization in multi-population mortality models. We adopt the concept introduced by the SVAR model to extract the optimal sparse structure from the data via the least absolute shrinkage and selection operator (LASSO) and improve it for multi-population mortality problems. Our innovation leverages the group properties of multi-population models and incorporates geographic information to improve model fitting. We introduce a spatial–temporal weight that considers age and geographical dimensions, integrating this weight into the sparse group penalty term in the estimation. This modification enhances the ability of LASSO to select non-zero parameters from similar age groups and geographically proximate populations. Empirical analysis demonstrates the effectiveness of this approach in terms of accuracy and stability.

We next outline the four main contributions of this paper. First, we comprehensively discuss existing single-population VAR models and identify their limitations in addressing multi-population problems. Second, we develop a grouped VAR model and a weighted sparse group LASSO estimation method tailored explicitly for multi-population mortality problems. Third, we demonstrate the effect of geographic information in multi-population mortality modeling, particularly in illustrating inter-population relationships. Finally, our SWVAR model surpasses the classical Li-Lee model, the coherent functional demographic model (Co-FDM), and other existing VAR models.

The remainder of the paper is organized as follows. Section 2 defines the relevant terms and notation and reviews classical mortality models and their limitations. Section 3 introduces the SWVAR model. Section 4 presents the empirical analysis, and Section 5 concludes the paper. The paper includes two appendices: the regular appendices contain the remaining details of this paper, while Online Resource 1 provides auxiliary figures related to Section 4.

2. Model formulation

2.1. Definitions and notations

Let us introduce the notations and settings used in this paper. Consider I populations with X age groups and data spanning T years. The mortality rate for the xth age group in population i during the year t is denoted as $m_{x,t,i}$. It is calculated as follows:

$$m_{x,t,i} = \frac{d_{x,t,i}}{e_{x,t,i}}, \tag{1}$$

where $d_{x,t,i}$ represents the observed number of deaths of people in the xth age group during year t in population i, and $e_{x,t,i}$ is the corresponding exposure to risk, calculated as the sum of the time being alive by all individuals within the xth age group during year t in population i[1]. Here, $i = 1, 2, \ldots, I$, $x = 1, 2, \ldots, X$, and $t = 0, 1, \ldots, T$, with the first year in the dataset corresponding to $t = 0$.

Vectors and matrices are written in bold throughout the remainder of the paper. The term log refers to the natural logarithm. Furthermore, we use "$\rightarrow$" to denote a "significant effect". For example, "$x_1 \rightarrow y$" indicates that the explanatory variable x_1 significantly affects the response variable y, suggesting that the corresponding regression coefficient may not be zero.

2.2. Existing mortality models

In this section, we present some popular methodologies for multi-population mortality modeling.

[1] For more details about the computation of $d_{x,t,i}$ and $e_{x,t,i}$, see Wilmoth et al. (2007).

2.2.1. The Li-Lee model and the coherent functional demographic model

The Li-Lee model (Li & Lee, 2005), also called the augmented common factor (ACF) model, extends the Lee–Carter model (Lee & Carter, 1992) to the multi-population setting. The Li-Lee model is defined in Appendix A.

One drawback of the Li-Lee model is that it fixes the bilinear structure between age and period effects and then forecasts population-specific factors with an AR(1) model to ensure that these factors converge to a constant value over time. This constraint limits the assessment of disparities among different populations to a narrow scope. In practice, this often results in poor forecast performance and degraded long-term predictions for some 'distinct' populations in a group. Among the 15 low-mortality populations studied by Li and Lee (2005), an example is Japan. It is geographically isolated from other countries and demographically distinct due to its aging population, extremely low fertility rates, and limited immigrant diversity. Another drawback of the Li-Lee model is that it can be considered overparameterized. For example, in Cupido et al. (2024), the Li-Lee model was fitted to the populations of 48 states in the United States of America; the model contains 3,222 parameters, while the data consists only of 48(number of populations) × 13(number of age groups) × 40(number of years) = 24,960 observations.

We discuss the coherent functional demographic model next. Using a functional data paradigm (Ramsay & Silverman, 2002), Hyndman and Ullah (2007) propose a non-parametric method to model and forecast log-mortality rates, namely, the Hyndman–Ullah (HU) method. To further ensure coherency in forecasting mortality rates for multiple populations based on these functional principal component models, Hyndman et al. (2013) introduce the coherent functional demographic model (Co-FDM), which proposes a product-ratio functional forecasting method. Co-FDM substantially extends the HU model, and we provide the definition in Appendix B.

2.2.2. The VAR model

Vector autoregressive models are introduced as a logical alternative method to factor-based models. It is popular because of its performance in modeling complicated structures and in forecasting.

Let $y_{x,t,i} = log(m_{x,t,i})$. Based on past observations, the log-mortality rate series $y_{x,t}$ is typically non-stationary. Therefore, we follow (Guibert et al., 2019) and use the first difference of the log-mortality rates, denoted by $z_{x,t} = y_{x,t} - y_{x,t-1}$, to remove the linear trend. This time series is generally shown to be stationary (Guibert et al., 2019). The series $z_{x,t}$ represents the log-mortality improvement rates. We apply a vector autoregressive (VAR) model to this stationary series. First, let us focus on the single-population setting: we fix population $i \in \{1, 2, \ldots, I\}$, and we will consider multi-population extensions in Section 2.2.5. The VAR(1) model is given by:

$$\mathbf{z}_t = \mathbf{c} + \mathbf{B}\mathbf{z}_{t-1} + \boldsymbol{\epsilon}_t, \tag{2}$$

where $\mathbf{z}_t = (z_{1,t}, z_{2,t}, \ldots, z_{X,t})'$, $\mathbf{c} = (c_1, c_2, \ldots, c_X)'$ is an X-dimensional intercept vector, and $\mathbf{B}$ is the coefficient matrix of dimension $X \times X$, also known as the Granger causality matrix (Granger, 1969). Here $\boldsymbol{\epsilon}_t$ is the vector of error terms, commonly assumed to be i.i.d. Gaussian with mean 0. Next, we introduce and discuss two pioneering approaches to VAR models.

2.2.3. Spatial–temporal autoregressive (STAR) model

Considering the model in (2), Li and Lu (2017) propose the STAR model, which accounts for sparsity and stationarity by constraining the shape of the Granger causality matrix as follows:

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 & \cdots & \cdots \\ \alpha_2 & 1-\alpha_2 & 0 & \cdots & \cdots \\ \beta_3 & \alpha_3 & 1-\alpha_3-\beta_3 & 0 & \cdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & \beta_X & \alpha_X & 1-\beta_X-\alpha_X \end{pmatrix}. \tag{3}$$

This hard restriction limits the interaction between non-neighbor age groups in a single population setting. Meanwhile, they add an L_2 regularization for $\mathbf{c}$ and a Frobenius norm for $\mathbf{B}$ to ensure their smoothness (see, e.g., Tsybakov, 2009). Specifically, the objective function of the SVAR(1) model is:

$$(\hat{\mathbf{c}}, \hat{\mathbf{B}}) = \arg\min_{\mathbf{c},\mathbf{B}} \frac{1}{2} \sum_{t=1}^{T} \|\mathbf{z}_t - \mathbf{c} + \mathbf{B}\mathbf{z}_{t-1}\|_2^2 + \eta_1 \left\|\mathbf{c}'\mathbf{H}\right\|_2^2 + \eta_2 \left\|\mathbf{B}'\mathbf{H}\right\|_F^2, \tag{4}$$

where for a d-dimensional vector $\mathbf{g} = (g_i)_{1\le i\le d} \in \mathbb{R}^d$, $\|\cdot\|_2^2$ is defined as:

$$\|\mathbf{g}\|_2^2 = \sum_{i=1}^{d} g_i^2, \tag{5}$$

and for a $d \times d$ matrix $\mathbf{G} = (g_{i,j})_{1\le i\le d, 1\le j\le d} \in \mathbb{R}^{d\times d}$, $\|\cdot\|_F^2$ is defined as:

$$\|\mathbf{G}\|_F^2 = \sum_{i=1}^{d}\sum_{j=1}^{d} g_{i,j}^2. \tag{6}$$

Additionally, the $\mathbf{H}$ matrix is defined as:

$$\mathbf{H} = \begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ -1 & 1 & \ddots & \vdots & \vdots \\ 0 & -1 & \ldots & 0 & 0 \\ \vdots & \ddots & \ddots & 1 & 0 \\ 0 & \cdots & 0 & -1 & 1 \end{pmatrix}, \tag{7}$$

and $\eta_1, \eta_2 \ge 0$ are hyperparameters selected by minimizing the mean square error (MSE).

2.2.4. Sparse vector autoregressive (SVAR) model

Instead of imposing an exogenous dependency structure, the SVAR model is another potential solution that has been extensively studied (see, e.g., Basu & Michailidis, 2015). Guibert et al. (2019) propose the SVAR model with an L_1 penalty to handle the issue of overparameterization in mortality modeling. This purely data-driven method

allows for a large and flexible VAR structure without excluding potentially relevant relationships.

The SVAR(1) model is given by:

$$\mathbf{z}_t = \mathbf{c} + \mathbf{B}\mathbf{z}_{t-1} + \boldsymbol{\epsilon}_t, \tag{8}$$

where $\mathbf{z}_t = (z_{1,t}, z_{2,t}, \ldots, z_{X,t})'$. Here, $\mathbf{c} = (c_1, c_2, \ldots, c_X)'$ is the intercept vector, and $\mathbf{B}$ is the coefficient matrix. Moreover, $\boldsymbol{\epsilon}_t$ is an i.i.d. Gaussian error vector with mean 0. The coefficients can then be estimated using an elastic net:

$$(\hat{\mathbf{c}}, \hat{\mathbf{B}}) = \arg\min_{\mathbf{c},\mathbf{B}} \frac{1}{2}\sum_{t=1}^{T} \|\mathbf{z}_t - \mathbf{c} + \mathbf{B}\mathbf{z}_{t-1}\|_2^2 + \lambda_1 \sum_{i=1}^{X}\sum_{j=1}^{X} |b_{i,j}| + \lambda_2 \sum_{i=1}^{X}\sum_{j=1}^{X} b_{i,j}^2, \tag{9}$$

where $(b_{i,j})_{1\le i\le X, 1\le j\le X}$ are the elements in coefficient matrix $\mathbf{B}$, and $\lambda_1, \lambda_2 \ge 0$ are hyperparameters, which are selected by minimizing the MSE.

2.2.5. Multi-population extension

Both the STAR and SVAR models primarily focus on single-population mortality modeling. While their direct extension to multi-population mortality modeling is feasible, they present some drawbacks.

The strength of the STAR model lies in using constraints to avoid overparameterization. The STAR model only allows correlations between neighboring age groups within the same population ($z_{x-1,t-1,i} \to z_{x,t,i}$ and $z_{x,t-1,i} \to z_{x,t,i}$) and correlations between the same age group across different populations ($z_{x,t-1,i_1} \to z_{x,t,i_2}$). For the two-population case, the model can be written as:

$$z_{x,t+1,i} = \rho_{i,-i,x}[(1-\alpha_{x,i})z_{x,t,i} + \alpha_{x,i}z_{x-1,t,i}] + (1-\rho_{i,-i,x})z_{x,t,-i} + c_{x,i} + \epsilon_{x,t+1,i}, \tag{10}$$

with $i = 1, 2$ and $-i = 1$ if $i = 2$ and vice versa, and $x > 1$. Under this specification, the multi-population extension has a sparse VAR representation:

$$\mathbf{B}_0 = \begin{pmatrix} 1 & 0 & 0 & \cdots & \cdots \\ \alpha_2 & 1-\alpha_2 & 0 & \cdots & \cdots \\ 0 & \alpha_3 & 1-\alpha_3 & 0 & \cdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & \alpha_X & 1-\alpha_X \end{pmatrix},$$

$$\mathbf{P}_{i,j} = \begin{pmatrix} \rho_{i,j,1} & 0 & \cdots & 0 \\ 0 & \rho_{i,j,2} & \cdots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ 0 & \cdots & 0 & \rho_{i,j,X} \end{pmatrix}, \tag{11}$$

$$\mathbf{z}_{t,i} = \mathbf{c}_i + \mathbf{P}_{i,i}\mathbf{B}_0\mathbf{z}_{t-1,i} + \sum_{j\neq i}\mathbf{P}_{i,j}\mathbf{I}_X\mathbf{z}_{t-1,j} = \mathbf{c}_i + (\mathbf{P}_{i,1}\mathbf{I}_X, \ldots, \mathbf{P}_{i,i}\mathbf{B}_0, \ldots, \mathbf{P}_{i,I}\mathbf{I}_X)\begin{pmatrix}\mathbf{z}_{t-1,1}\\ \vdots \\ \mathbf{z}_{t-1,i}\\ \vdots \\ \mathbf{z}_{t-1,I}\end{pmatrix}, \tag{12}$$

where I_X is the identity matrix of size X and $\mathbf{c}_i$ is the mean vector that corresponds to population i. From (12), we can see the STAR model imposes a more restrictive structure to the Granger matrix, permitting only diagonal coefficients (and the lower subdiagonal coefficients of the corresponding population) to be non-zero. This constraint will lead to overlooking various relationships, including the cross-population cohort effect.

On the other hand, in the SVAR model, we stack all populations' mortality data into a vector:

$$\mathbf{z}_t = \begin{pmatrix}\mathbf{z}_{t,1}\\ \mathbf{z}_{t,2}\\ \vdots \\ \mathbf{z}_{t,I}\end{pmatrix}, \tag{13}$$

where $\mathbf{z}_{t,1} = (z_{1,t,1}, z_{2,t,1}, \ldots, z_{X,t,1})'$. In the multi-population SVAR model, the formula (8) remains unchanged. The coefficient matrix is still estimated by solving (9), with the only distinction being that the dimension of $\mathbf{z}_t$ transitions from X to IX. However, the fitting results produce parameter estimates that are hard to relate to reality and actuarial theory for multi-population mortality modeling. The main issue is the model's inability to distinguish between different age groups and populations. This follows from the model assumption that the model takes the reduction in the score function as the sole criterion for selecting non-zero parameters. This presents a significant challenge, as reductions in the score function may be due to coincidental factors specific to certain data points rather than reflecting the underlying "true" patterns. Therefore, the following section proposes a more interpretable and sparse multi-population VAR model.

3. Our multi-population model

In this section, we specify the proposed SWVAR model. To eliminate the intercept term throughout the SWVAR model, we centralize $\mathbf{z}_{x,t,i}$ in (13) for all $x = 1, \ldots, X$ and $i = 1, \ldots, I$, so that the observed mean is $\mathbf{0}$.

3.1. Multi-population VAR

The stacked multi-population VAR(1) model can be written as:

$$\mathbf{z_t} = \begin{pmatrix}\mathbf{z}_{t,1}\\ \mathbf{z}_{t,2}\\ \vdots \\ \mathbf{z}_{t,I}\end{pmatrix} = \begin{pmatrix}\mathbf{B}_{1,1} & \cdots & \mathbf{B}_{1,I}\\ \vdots & \ddots & \vdots \\ \mathbf{B}_{I,1} & \cdots & \mathbf{B}_{I,I}\end{pmatrix}\begin{pmatrix}\mathbf{z}_{t-1,1}\\ \mathbf{z}_{t-1,2}\\ \vdots \\ \mathbf{z}_{t-1,I}\end{pmatrix} + \begin{pmatrix}\boldsymbol{\epsilon}_{1,t}\\ \boldsymbol{\epsilon}_{2,t}\\ \vdots \\ \boldsymbol{\epsilon}_{I,t}\end{pmatrix} = \mathbf{B}\mathbf{z}_{t-1} + \boldsymbol{\epsilon}_t, \tag{14}$$

where $\mathbf{z}_{t,1} = (z_{1,t,1}, z_{2,t,1}, \ldots, z_{X,t,1})'$. Here, $\mathbf{z}_t$ is an IX-dimensional vector (IX stands for $I \times X$). Here, $\mathbf{B}$ is the coefficient matrix with dimension $IX \times IX$ and $\mathbf{B}_{i,j}$ is the block in the ith row and jth column of $\mathbf{B}$ with dimension

$X \times X$. Moreover, $\boldsymbol{\epsilon}_t$ is an i.i.d. Gaussian error vector. We can treat $\mathbf{z}_{t,1}$ as a set of variables in this autoregressive model. Then, for the multi-population VAR, the vector $\mathbf{z}_t$ can be divided into I different groups, each containing X variables.

3.2. Sparse group Lasso

As discussed in the Introduction, our primary concern is overparameterization. The number of coefficients for a single-population VAR(1) model is X^2. If we stack I populations into one vector, the number of coefficients becomes $(IX)^2$, while generally less than $T = 100$ years of historical data are used for estimation, where $(IX)^2 \gg T$. Although the SVAR and STAR models attempt to address overparameterization, their strategies are effective mainly in single-population models, and we show that their performance is worse in a multi-population model.

Due to the inefficiency of existing methods, a more effective variable selection method is needed. According to the grouped property of the multi-population VAR model, the selection of important variables corresponds to the selection of groups. The group LASSO (Yuan & Lin, 2006) is introduced to address this problem. For a vector $\mathbf{g} \in \mathbb{R}^d$, $d \geq 1$, and a symmetric $d \times d$ positive definite matrix $\mathbf{K}$, we denote:

$$\|\mathbf{g}\|_K = (\mathbf{g}'\mathbf{K}\mathbf{g})^{1/2}. \tag{15}$$

Since the regression coefficients of a VAR model consist of a matrix, we need to extend (15) to a matrix $\mathbf{G} \in \mathbb{R}^{d\times d}$ with:

$$\|\mathbf{G}\|_K = \sum_{i=1}^{d} (\mathbf{g}_{i,}'\mathbf{K}\mathbf{g}_{i,})^{1/2}, \tag{16}$$

where $\mathbf{g}_{i,}$ denotes the ith row of the matrix $\mathbf{G}$. Then, given positive definite matrices $(\mathbf{K}_{i,j})_{1\leq i\leq I, 1\leq j\leq I}$, the group LASSO estimation for the multi-population mortality VAR model can be defined as:

$$\hat{\mathbf{B}} = \arg\min_{\mathbf{B}} \sum_{i=1}^{I}\sum_{t=1}^{T} \left\| \mathbf{z}_{t,i} - \sum_{j=1}^{I} \mathbf{B}_{i,j}\mathbf{z}_{t-1,j} \right\|_2^2 + \lambda \sum_{i=1}^{I}\sum_{j=1}^{I} \left\|\mathbf{B}_{i,j}\right\|_{\mathbf{K}_{i,j}}, \tag{17}$$

where $\mathbf{B}_{i,j}$ is defined in Section 3.1, and $\lambda \geq 0$ is a hyperparameter, which is selected via a minimization of the MSE.

However, when the group LASSO includes a group in the model, then all coefficients in this group will be non-zero. This is not ideal because we do not expect a complete in-group correlation structure. For example, a strong effect from the newborn group to older age groups is not realistic, even in a single-population VAR model. We seek both sparsity of groups and sparsity within each group; hence, we propose the sparse group LASSO (Simon et al., 2013):

$$\hat{\mathbf{B}} = \arg\min_{\mathbf{B}} \sum_{t=1}^{T}\sum_{i=1}^{I} \left\| \mathbf{z}_{t,i} - \sum_{j=1}^{I} \mathbf{B}_{i,j}\mathbf{z}_{t-1,j} \right\|_2^2 + (1-\alpha)\lambda \sum_{i=1}^{I}\sum_{j=1}^{I} \left\|\mathbf{B}_{i,j}\right\|_{\mathbf{K}_{i,j}} + \alpha\lambda \sum_{i=1}^{IX}\sum_{j=1}^{IX} \left|b_{i,j}\right|, \tag{18}$$

where $\|\cdot\|_2^2$ is defined in (5), and $\|\cdot\|_{\mathbf{K}_{i,j}}$ is defined in (16). We use $(b_{i,j})_{1\leq i\leq IX, 1\leq j\leq IX}$ to represent the elements in $\mathbf{B}$, and $\alpha \in [0, 1]$ and $\lambda \geq 0$ are hyperparameters.

3.3. Spatial–temporal weight

In addition to a group structure, useful prior information can be used in multi-population mortality modeling. The age group can be seen as a temporal gap, and the geographic distance contains spatial information. In this section, we create an adaptive spatial–temporal weight matrix, denoted by **Wst**, to help select variables more reasonably.

We use an age-adaptive weight in the SWVAR model to describe in-group relations. The age weight matrix is denoted by $\mathbf{Wa} \in \mathbb{R}^{X\times X}$. The coefficient between age groups i and j should be more penalized for larger $|i-j|$, and thus the corresponding coefficients are more likely to shrink toward zero:

$$wa_{i,j} = \exp\left\{\frac{|i-j|}{\theta}\right\}, \tag{19}$$

where θ is a non-negative tuning parameter that controls the sensitivity of shrinkage to the distance between populations i and j. This age-adaptive weight is also used by Chang and Shi (2021) for single-population mortality modeling,

Regarding inter-population relationships, we resort to geographic information in the spirit of Quick et al. (2018) and Cupido et al. (2024). Instead of using a $0-1$ adjacency matrix used by those two papers, we use the centroid distance as an approximation. This assumption is based on the hypothesis that populations closer to each other are naturally more correlated, and populations further apart may still be correlated. Then, the adaptive spatial weight matrix, denoted by $\mathbf{Ws} = (ws_{i,j}) \in \mathbb{R}^{I\times I}$, is given by:

$$ws_{i,j} = \begin{cases} 1 & \text{if } i = j, \\ \gamma \times \log(d(i,j)) & \text{otherwise,} \end{cases} \tag{20}$$

where $d(i,j)$ is the centroid distance between population i and j, and γ is the non-negative spatial tuning parameter, which controls how much other populations influence a population. Then, the final adaptive spatial–temporal weight matrix, denoted as $\mathbf{Wst} \in \mathbb{R}^{IX\times IX}$, is given by:

$$\mathbf{Wst} = \mathbf{Ws} \otimes \mathbf{Wa}, \tag{21}$$

where $\otimes$ is the Kronecker product.

3.4. Estimation of the SWVAR model

To fully utilize the prior information during the LASSO estimation, we first separate the problem in (18) by population:

$$\mathbf{z}_{t,i} = \begin{pmatrix} \mathbf{B}_{i,1} & \cdots & \mathbf{B}_{i,I} \end{pmatrix} \begin{pmatrix} \mathbf{z}_{t-1,1} \\ \mathbf{z}_{t-1,2} \\ \vdots \\ \mathbf{z}_{t-1,I} \end{pmatrix} + \boldsymbol{\epsilon}_{t,i} = \sum_{j=1}^{I} \mathbf{B}_{i,j}\mathbf{z}_{t-1,j} + \boldsymbol{\epsilon}_{t,i}, \tag{22}$$

where $i = 1, 2, \ldots, I$. Then the *spatial–temporally weighted sparse group LASSO* for one population is:

$$\hat{\mathbf{B}}_{i,\cdot} = \arg\min_{\mathbf{B}_{i,\cdot}} \sum_{t=1}^{T} \left\| \mathbf{z}_{t,i} - \sum_{j=1}^{I} \mathbf{B}_{i,j}\mathbf{z}_{t-1,j} \right\|_2^2 + (1-\alpha)\lambda \sum_{j=1}^{I} \left\| \mathbf{B}_{i,j} \right\|_{\mathbf{K}_{i,j}} + \alpha\lambda \sum_{j=1}^{I} wst_{i,j} \sum_{m=1}^{X}\sum_{n=1}^{X} |b_{im,jn}|, \tag{23}$$

where $\mathbf{B}_{i,\cdot}$ means all block-elements in the ith row of $\mathbf{B}$ and $b_{im,jn}$ means the element in the mth row and nth column of $\mathbf{B}_{i,j}$. Furthermore, the matrix $\mathbf{K}_{i,j}$ for the group penalty term follows:

$$\mathbf{K}_{i,j} = ws_{i,j} \cdot \mathbf{I}_X. \tag{24}$$

Then we assume the error term $\boldsymbol{\epsilon}_{t,i}$ follows i.i.d multivariate Gaussian distribution: $\boldsymbol{\epsilon}_{t,i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \boldsymbol{\Sigma}_i)$, so different population i have different $\boldsymbol{\Sigma}_i$.

Four hyperparameters must be selected jointly. We select them jointly through two steps: an "inner loop" and an "outer loop". In the "inner loop", we first select best LASSO parameters via cross-validation to choose the α and λ with the least MSE.[2] Then, in the "outer loop", we choose θ and γ according to an information criterion. We first derive the log-likelihood function $l(\hat{\mathbf{B}}, \hat{\boldsymbol{\Sigma}})$ of our SWVAR model:

$$l(\hat{\mathbf{B}}, \hat{\boldsymbol{\Sigma}}) = \sum_{i=1}^{I} \left(-\frac{XT}{2}\log 2\pi - \frac{T}{2} log|\hat{\boldsymbol{\Sigma}}_i| - \frac{1}{2}\sum_{t=1}^{T} (\mathbf{z}_{t,i} - \hat{\mathbf{z}}_{t,i})' \hat{\boldsymbol{\Sigma}}_i^{-1} (\mathbf{z}_{t,i} - \hat{\mathbf{z}}_{t,i}) \right), \tag{25}$$

where $\hat{\boldsymbol{\Sigma}}$ is the empirical covariance matrix, $\hat{\mathbf{B}}$ is the solution of (23) and $\hat{\mathbf{z}}_{t,i} = \hat{\mathbf{B}}\mathbf{z}_{t-1,i}$ is the estimator of $\mathbf{z}_{t,i}$ under the SWVAR model. The AIC and BIC are defined as follows:

$$\begin{aligned} \text{AIC} &= 2 \times l(\hat{\mathbf{B}}, \hat{\boldsymbol{\Sigma}}) - 2 \times edf, \\ \text{BIC} &= 2 \times l(\hat{\mathbf{B}}, \hat{\boldsymbol{\Sigma}}) - log(T) \times edf, \end{aligned} \tag{26}$$

where edf is the effective degree of freedom, equal to the number of non-zero coefficients in $\mathbf{B}$. Since higher AIC/BIC indicates better model fitting, we will do a 2-dimensional grid search to select the spatial–temporal hyperparameters θ and γ with the highest AIC/BIC.[3] In conclusion, the estimation process of our SWVAR model follows Algorithm 1.

Algorithm 1 SWVAR

Require: Predetermined grid search list for θ and γ, denote by l_θ and l_γ.
Ensure: Estimated coefficient matrix $\mathbf{B}$.

```
for i = 1 to I do
    Take z_i as the variable and solve (23).
    for θ in l_θ do
        for γ in l_γ do
            Compute the spatial-temporal weight matrix.
            Select best LASSO parameters α, λ via cross validation with minimum MSE.
            Compute resulting AIC (BIC) value by (26).
        end for
    end for
    Select best θ and γ with minimum AIC (BIC) for population i, get the corresponding estimated B̂_{i,·}.
end for
```

Since other populations influence each population differently in terms of mortality, and the relationship between different age groups varies within populations, separate single-population modeling offers an advantage. Specifically, it allows us to identify the optimal spatial–temporal parameters for each population individually.

3.5. Model selection

After introducing the SWVAR model, our objective is to evaluate its feasibility and performance more thoroughly. Although AIC and BIC are commonly used for model selection by balancing fit and parameter count, they are limited in their ability to differentiate between models with similar values. Consequently, these criteria cannot comprehensively evaluate the effectiveness of the SWVAR model or determine if it is superior to other VAR models.

Therefore, we focus on more nuanced and interpretable model comparison mechanisms. It is important to note that both the SVAR and SWVAR models aim to fit a sparse VAR model, each seeking the optimal non-zero parameters under different criteria. Specifically, the SWVAR model incorporates an additional spatial–temporal weight on the penalized term compared to the SVAR model. Comparing two models with different structures fits well with the task of non-nested model comparison. Therefore,

[2] We wrote our code based on the R package **sparsevar** (Vazzoler, 2021), which used the widely used the R package **glmnet** (Friedman et al., 2010) to solve the penalized estimation. In our empirical results, we also use **glmnet** and modify it according to the objective functions in the SVAR and SWVAR models. If only the SWVAR model is considered, the R package **sparsegl** (Liang et al., 2024) can be an alternative for **glmnet**.

[3] In our empirical results, we use the AIC as the selection criterion.

we perform two non-nested model comparison tests, the C-test and the J-test (see Pesaran & Deaton, 1978, and Davidson & MacKinnon, 1981), for our evaluation. Detailed definitions and explanations of the C-test and J-test are provided in Appendix C.

4. Empirical analysis

In this section, we apply the SWVAR model introduced in the previous section to historical mortality data and compare it with the Li-Lee model (Li & Lee, 2005), the Co-FDM model (Hyndman et al., 2013), the STAR model (Li & Lu, 2017), the unconstrained VAR model (with direct least-squares estimation, denoted by VAR), and the SVAR model (Guibert et al., 2019). We explore two different approaches for both the STAR and the SVAR models. The first approach is single-population modeling, where we apply the models described in Sections 2.2.3 and 2.2.4 to each population. We refer to this approach as the single-population SVAR/STAR model or 'SVAR/STAR single' for short. The second approach uses the stacked method for multi-population modeling introduced in Section 2.2.5, denoted as the stacked SVAR/STAR model, or 'SVAR/STAR stack' for short. In the detailed analysis, we do not focus on the difference between the stacked and single-population STAR and SVAR models; we only compare the stacked model with other multi-population models. Therefore, we omit the word 'stack' in figures and tables for conciseness if only one SVAR and STAR model is presented.

We apply the VAR, STAR, SVAR, and SWVAR models to the first differences of the log-mortality rate series. We omit high-order lags to reduce the risk of overparameterization. Furthermore, the in-sample performance of the VAR(1) model, as measured by AIC/BIC, is superior to that of higher-order VAR models.

4.1. Data

Following Li and Lee (2005), we select three groups for empirical studies. Group 1 consists of 15 low-mortality countries, excluding West Germany, with the sample period from 1950 to 2019. Group 2 comprises six Eastern European countries, excluding East Germany; the sample period is from 1959 to 2013. Group 3 consists of six former Soviet Union countries, with the same sample period as Group 2. For a more comprehensive evaluation, we add Group 4, which combines all populations in Group 2 and Group 3, to achieve a larger sample size. In contrast, Group 5, which includes smaller populations with adjacent spatial relationships, is selected from Group 1. The populations in the five groups are listed in Table 6 in Appendix D. Our data are obtained from the Human Mortality Database.[4] In this section, we fit all models with male data; female data are used in Section 4.5.2 to verify the robustness of the model.

We next discuss the choice of age groups. The original method employs five-year age groups for the Li-Lee model: $0, 1-4, 5-9, \ldots, 95-99$. In contrast, many VAR models use one-year age groups to provide a more detailed view of the dynamic interdependencies (e.g., Chang & Shi, 2021; Guibert et al., 2019; Li & Lu, 2017). However, the prediction performance of the stacked STAR model is poor for older ages when fitted with one-year age groups. Furthermore, the number of coefficients for a VAR model with 15 populations and one-year age groups $0, 1, \ldots, 99$ is $(100 \times 15)^2 = 2,250,000$, which is substantially more compared to $(21 \times 15)^2 = 99{,}225$ for a model with five-year age groups. For both the SVAR and SWVAR models, which rely on iterative optimization, a model with five-year age groups is much more efficient and stable. Consequently, all models are fitted using five-year age group data.

4.2. Computational packages

We use the software R to perform the computations of all models. The Li-Lee model is fitted by the **MortalityForecast** (Pascariu, 2020) package. Co-FDM is estimated using the **vital** (Hyndman, 2024) package. The SVAR model is implemented via the **sparsevar** (Vazzoler, 2021) package. The STAR and SWVAR models are estimated using code written by the authors. We use the **CVXR** (Wang et al., 2020) package to solve the constrained optimization problem within the STAR model. We use both R packages **glmnet** (Friedman et al., 2010) and **sparsegl** (Liang et al., 2024) to solve for the penalized estimation in the SWVAR model.

4.3. Estimation

In this section, we focus on studying the estimated Granger causality matrices that describe the long-term underlying mortality dynamic of the model. Specifically, we first report the estimation results of the SWVAR model and compare them with the results of the stacked SVAR model. Then, we present the demographic implications inferred from the estimation results.

A part of the coefficient matrix $\mathbf{B}$ from (14) for Group 1 is shown in Fig. 1. We select two subgroups of Group 1 for illustration; the rest are reported in Online Resource 1 (Figs. S1, S6, S11).[5] The estimated matrix $\mathbf{B}$ is sparse, meaning that most of the coefficients are estimated to be 0 when minimizing the criterion given in (9) and (22).

In a single-population SVAR model, three types of effects are considered (Guibert et al., 2019): 1) cohort effect, 2) period effect, and 3) age-specific effect with vertical structure. In our empirical results, since we use five-year age group data, the cohort effect ($y_{x-1,t-1} \rightarrow y_{x,t}$) is mixed with the period effect ($y_{x,t-1} \rightarrow y_{x,t}$). The mixing of cohort and period effects is presented by coefficients on the diagonal of B. The vertical structure persists, indicating that mortality improvements in older age groups are influenced by middle age groups, which may reflect specific social effects.

[4] The Human Mortality Database is available at: http://www.mortality.org.

[5] For comparison, we display the parameter estimates of the Li-Lee model for Group 1 in Fig. 8 in Appendix A.

When we extend to the multi-population scenario, we separately analyze two structures by observing the coefficients in B: within-population and inter-population effects. The within-population effects were observed on the diagonal submatrices. Large negative values for almost all young groups and some old groups in the diagonal of diagonal submatrices indicate significant cohort and period effects. Meanwhile, most middle-aged groups have zero estimates, showing that they are less autocorrelated between age groups and more stable with respect to the time dimension. On the other hand, the inter-population effect, representing the cross-correlations in the multi-population mortality model, is presented by coefficients in the off-diagonal matrices of B.

The key distinction between the SWVAR and SVAR models lies in selecting non-zero coefficients. The SVAR model, as a purely data-driven method, performs well in single-population models but tends to produce ambiguous coefficients in multi-population scenarios due to its inability to differentiate between populations and age groups within populations. In contrast, the Group LASSO estimation enables the SWVAR model to select non-zero coefficients by population, and the SWVAR model uses the spatial–temporal weight W_{ST} as described in Section 3.3, which ensures that non-zero coefficients are more likely to be found among local populations and adjacent age groups. When $|x_1 - x_2|$ is large and population i_1 is distant from population i_2, the corresponding coefficient is more likely to be penalized to zero, aligning with actuarial and demographic theory. As a result, our SWVAR model significantly reduces the irregular and uninterpretable within-population and inter-population effects.

As shown in Fig. 1(a), the coefficient matrix of the SVAR model exhibits a random pattern that indicates correlations between almost every pair of populations. For example, the improvement in mortality in young age groups from Austria appears to influence the improvement in mortality in young and middle-aged groups from Norway and Sweden. While not necessarily incorrect, this outcome resembles noise, as mortality improvements at young ages should not exhibit strong inter-population correlations with other populations' middle-aged groups. Although this estimation may decrease the loss function in fitting, it may adversely impact mortality forecasting for Norway and Sweden due to these unexplainable inter-population effects. Furthermore, Fig. 1(b) shows that the SVAR model produces an extremely sparse estimation, with no apparent Granger correlation between Canada and the USA or between England and France, highlighting the SVAR model's inability to capture the characteristics of different populations. This can be seen as local underfitting, leading to stable but poor forecasts.

In contrast, the non-zero fitted coefficients from the SWVAR model are concentrated on the diagonals and blocks where corresponding populations are generally highly correlated (e.g., Canada and the USA, England and France). This result aligns with the expectation that spatially close groups are more likely to be related or exhibit significant effects on each other. Compared to the SVAR model, the SWVAR model reduces randomness in parameters across all blocks, yielding more reliable results. For example, the colored area in the fourth row and third column of SWVAR, and its counterpart in the third row and fourth column in Fig. 1(a), indicate that the mortality rates of Sweden are correlated with those of Norway. Similarly, the colored section in the first row and fourth column of SWVAR in Fig. 1(b) shows that the mortality rates of the USA help in modeling those of Canada. These results indicate that the proposed SWVAR model can selectively utilize data from other subpopulations to create better estimations. Consequently, mortality rates in more correlated populations will be forecasted jointly and reasonably. For example, this model combines the mortality rate of the younger age group in Canada this year with that of the same age group in the United States to predict the mortality rate of the younger age group in Canada. It does this without considering the mortality rates of any age group in England.

Interestingly, the non-zero effect is significantly weaker and more dispersed in the symmetrically reversed position (from Canada to the USA). Meanwhile, some populations exhibit a sparse structure, with most age groups estimated as constants only. In contrast, others display complex between-group relationships. These observations suggest that different populations are influenced differently by others, supporting the notion that the Li-Lee model and the predetermined ad hoc structure of the SVAR model may not be universally suitable.

4.4. In-sample analysis

4.4.1. General in-sample comparison

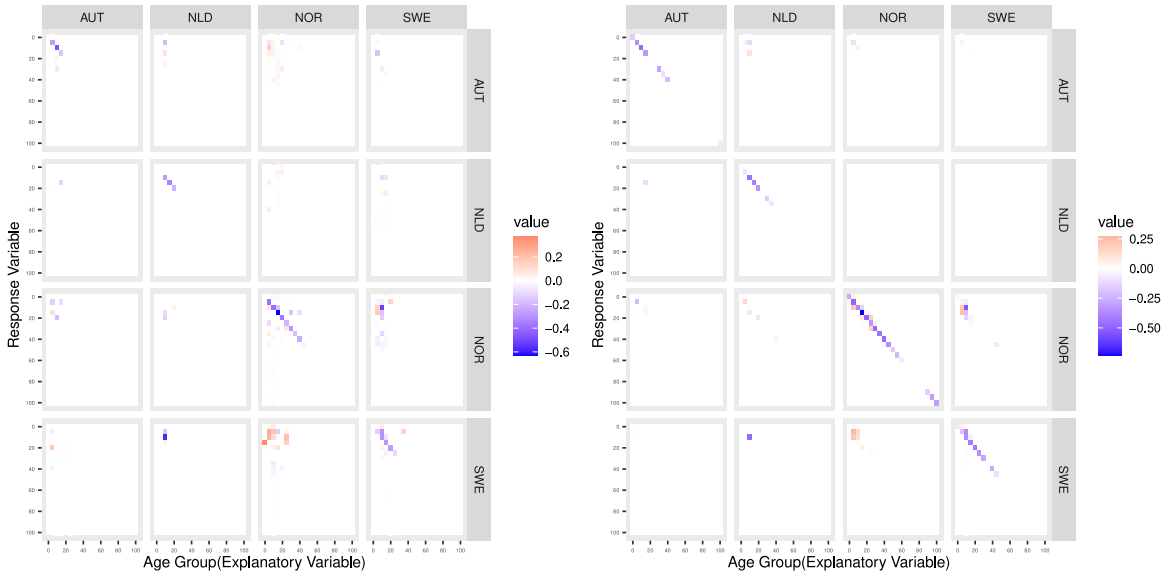
We compare the in-sample performance of all the mortality models discussed in this paper. A widely used evaluation metric is the root mean square error (RMSE) for all training data:

$$\text{RMSE} = \sqrt{\frac{1}{I \times X \times T} \sum_{t=1}^{T} \sum_{x=1}^{X} \sum_{i=1}^{I} (y_{x,t,i} - \hat{y}_{x,t,i})^2}, \tag{27}$$

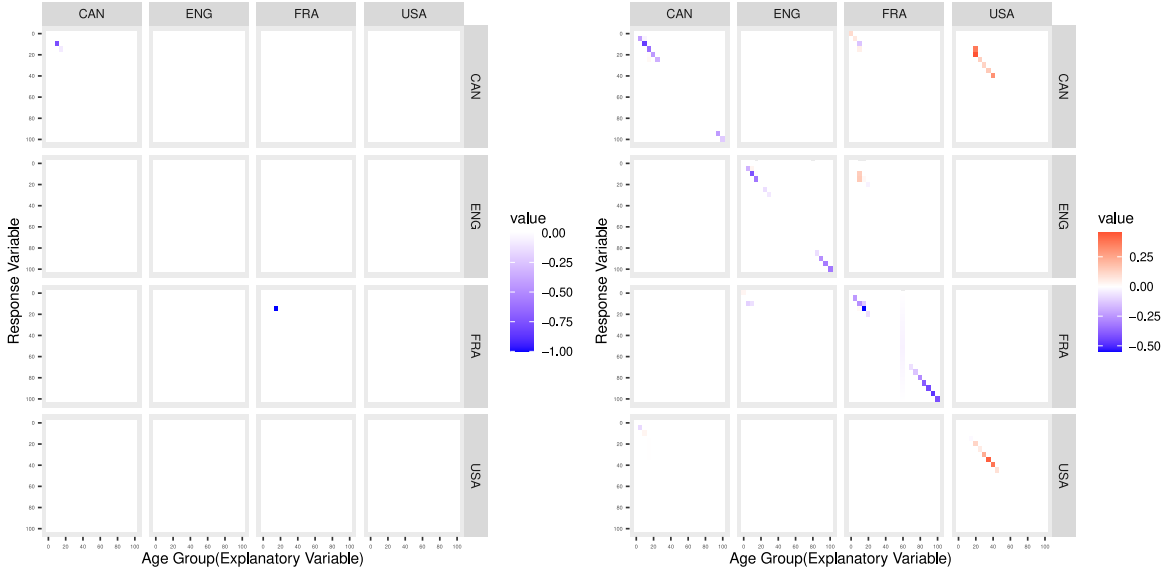
where $y_{x,t,i}$ is the observed log-mortality rate and $\hat{y}_{x,t,i}$ is the corresponding estimated value. The SWVAR model applies a VAR model to log-mortality improvements, so the estimated log-mortality rates are $\hat{y}_{x,t+1,i} = \hat{z}_{x,t,i} + y_{x,t,i}$, where $\hat{z}_{x,t,i}$ is the estimated value of $z_{x,t,i}$. We have

$$\begin{aligned} y_{x,t,i} - \hat{y}_{x,t,i} &= y_{x,t,i} - (y_{x,t-1,i} + \hat{z}_{x,t,i}) \\ &= y_{x,t,i} - y_{x,t-1,i} - \hat{z}_{x,t,i} = z_{x,t,i} - \hat{z}_{x,t,i}. \end{aligned}$$

So, for the RMSE calculation, there is no difference between considering the log mortality rates (y) and their first differences (z). In the log-likelihood calculations, we unify all models to the scale of log-mortality rates. Table 1 displays the effective degree of freedom (*edf*), equal to the number of non-zero parameters and the other previously mentioned evaluation metrics. The raw VAR model has the lowest RMSE because it has the largest number of non-zero parameters. The stacked model achieves better fitting results with more non-zero coefficients among the two STAR models. This improvement



(a) Four European Populations



(b) Two Neighboring Pairs

Fig. 1. Part of the estimated coefficient matrix **B** for the SVAR model (left) and the SWVAR model (right). The blank areas indicate that the corresponding coefficients are estimated to shrink to zero, while the colored blocks indicate non-zero estimates. Note that the scales of the color legends in the four figures differ since they represent different parts of **B**, but this does not affect our analysis.

is intuitive, as the joint modeling of different populations enhances the model's performance. In contrast, the stacked SVAR model does not show significant improvements compared to the single-population SVAR model. This suggests that the stacked SVAR model may not effectively utilize cross-group information. Our SWVAR model exhibits the highest log-likelihood across all groups except for Group 3. And with nearly the same *edf*, the SWVAR model has higher AIC and BIC values than the Li-Lee, STAR, and SVAR models, indicating a more accurate fitting. The exception is Group 3, where the stacked SVAR model shows higher log-likelihood and BIC than the SWVAR model, which may be attributed to the spatial weight. The large difference in territory may introduce a more complex spatial structure not fully captured by the log centroid distance.

Table 1
In-sample analysis.

Group	Metric	Li-Lee	Co-FDM	STAR stack	STAR single	VAR	SVAR stack	SVAR single	SWVAR
G1	RMSE	0.0957	0.0651	0.0634	0.0778	0.0399	0.0672	0.076	0.0726
	edf	1644	8114	4102	840	6527	569	571	682
	logL	−19696	−11331	−11469	−12620	−6973	−12960	−12838	−11851
	AIC	−42681	−38890	−31143	−26921	−27000	−27058	−26817	**−25066**
	BIC	−46354	−57018	−40307	−28797	−41582	−28329	−28093	**−26590**
G2	RMSE	0.0929	0.0631	0.0718	0.0807	0.0347	0.0752	0.0795	0.0748
	edf	567	2777	520	300	3489	217	197	269
	logL	−4803	−3331	−3621	−3756	−1116	−3846	−3855	−3556
	AIC	−10739	−12216	−8282	−8112	−9210	−8127	−8104	**−7649**
	BIC	−11888	−17840	−9335	−8719	−16277	−8566	−8503	**−8194**
G3	RMSE	0.0914	0.0625	0.0919	0.1016	0.065	0.0964	0.0981	0.0997
	edf	658	3198	750	360	1918	154	236	152
	logL	−5226	−4200	−6812	−5299	−3078	−4995	−5136	−5228
	AIC	−11768	−14797	−15124	−11319	**−9992**	−10299	−10744	−10760
	BIC	−13089	−21216	−16629	−12041	−13842	**−10608**	−11217	−11065
G4	RMSE	0.0947	0.0633	0.0812	0.0935	0.0561	0.0872	0.0907	0.0888
	edf	949	4569	1692	540	3153	294	367	315
	logL	−7930	−6143	−7780	−7591	−4312	−7583	−7359	−6828
	AIC	−17758	−21424	−18944	−16262	−14930	−15753	−15452	**−14286**
	BIC	−19663	−30595	−22340	−17346	−21259	−16344	−16188	**−14918**
G5	RMSE	0.0906	0.0635	0.0602	0.0685	0.0422	0.0644	0.0696	0.0688
	edf	756	3786	750	360	2404	128	191	191
	logL	−10394	−4517	−5426	−5361	−3252	−5627	−5551	−5187
	AIC	−22299	−16606	−12353	−11442	−11312	−11511	−11483	**−10756**
	BIC	−23988	−25064	−14028	−12246	−16683	−11797	−11910	**−11183**

Note: logL is short for the log-likelihood. Boldfaced and red values are the highest among all models.

Table 2
C-test.

Group	Estimation	Std	t statistic	p value
Group 1	1.00294	0.02490	40.28476	<2e−16*
Group 2	0.67266	0.05417	12.41745	<2e−16*
Group 3	1.05750	0.04460	23.71285	<2e−16*
Group 4	−0.15768	0.03176	–	–
Group 5	0.49513	0.05146	9.62082	<2e−16*

Note: The stars after the p-values indicate significance at the level 0.01. Then, we can reject the null hypothesis with a 99%-confidence level and conclude our SWVAR model is effective in model fit compared to the SVAR model. The C-test failed in Group 4 for the estimation is negative.

Table 3
J-test.

Group	Λ statistic	p-value
Group 1	864	<2.2e−16*
Group 2	380	<2.2e−16*
Group 3	104	<2.2e−16*
Group 4	289	<2.2e−16*
Group 5	342	<2.2e−16*

Note: The stars after the p-values indicate significance at the level 0.01. Then, we can reject the null hypothesis with a 99%-confidence level and conclude that our SWVAR model improves the model fit when compared to the unconstraint VAR model.

4.4.2. Model selection tests

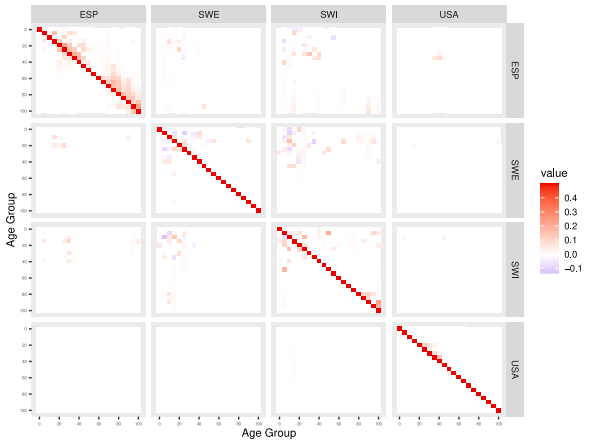
Incorporating the spatial–temporal weight into the LASSO estimation enhances the fit of the multi-population mortality model to the data. This adjustment slightly increases the number of non-zero coefficients but causes a substantial change in the model structure. Given the limitations of AIC/BIC in evaluating the model effectiveness, we conduct two tests introduced in Section 3.5. Both the C-test (Table 2) and J-test (Table 3) indicate that the SWVAR model performs well for multi-population mortality modeling. Consequently, the spatial–temporal weight is effective for multi-population modeling. The exception observed in Group 4 for the C-test also suggests that the centroid distance may be inadequate when the geographical structure of the populations is excessively complex with some very large distances.

4.4.3. Residual analysis

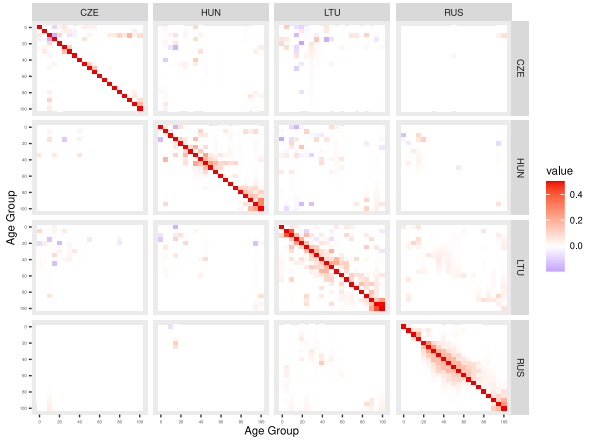
Finally, we compare the residuals produced by the stacked SVAR and SWVAR models. A fundamental step for regression models is to check the structure of the residuals. In our empirical findings, we expect the vector of residuals of different populations to be independent. We use a graphical LASSO (Friedman et al., 2008) under a Gaussian assumption to estimate the covariance matrix of residuals and transform it into a partial correlation matrix. Consider a set of variables $\mathbf{V} = \{\mathbf{y}_1, \ldots, \mathbf{y}_l\}$. If the corresponding precision matrix is defined as $\mathbf{P} = (p_{i,j}) = \boldsymbol{\Sigma}^{-1}$, then the partial correlation can be derived as follows:

$$\rho_{\mathbf{y}_i\mathbf{y}_j\cdot\mathbf{V}\setminus\{\mathbf{y}_i,\mathbf{y}_j\}} = -\frac{p_{i,j}}{\sqrt{p_{i,i}p_{j,j}}}, \tag{28}$$

where $\mathbf{y}_i, \mathbf{y}_j \in \mathbf{V}$.



(a) Group 1



(b) Group 2

Fig. 2. A portion of the partial correlation matrix is shown here. The upper triangle shows the partial correlation of residuals from the SVAR model, while the lower triangle shows those from the SWVAR model. We adjust the values on the diagonal to 0.5 for a better visual effect.

Fig. 2 displays part of the partial correlation matrix of residuals from the SVAR and SWVAR models for Groups 1 and 2. The remainder is reported in Online Resource 1 (Figs. S2, S7, S12). As shown in Fig. 2, the partial correlation of residuals from the SVAR model exhibits a more complex structure. The vertical and horizontal lines at the edges of some blocks indicate violations of the model assumptions. In contrast, our SWVAR model outperforms in terms of sparsity and reasonableness in most intra-population correlation blocks. For the first two blocks in the fourth row of Fig. 2(a), there are no partial correlations for the group as a whole. Furthermore, the SWVAR model effectively reduces inner-population correlations. For example, the block in the first column and first row in Fig. 2(a) and the block in the third column and third row in Fig. 2(b) show sparser inner-population correlation structures for Spain and Lithuania.

4.5. Out-of-sample performance

4.5.1. General out-of-sample comparison

We next compare the out-of-sample forecast performance of the SWVAR model with the mortality models defined in Section 2. First, we split the data from 1950 to 1999 as the training set and use the remaining part as the test set. Since Group 4 fails to reject the null in the C-test (see Table 2), we only empirically study the forecasts for the remaining four groups. We use the root mean square *forecasting* error (RMSFE) and mean absolute error (MAFE) to compare the forecasting performance of the mortality models. Four different RMSFE calculations are used in the comparison from various perspectives: overall $RMSFE_{all}$, $RMSFE_{pop}$ for specific populations, $RMSFE_{all,h}$ for overall populations at a given period on the forecast horizon, and $RMSFE_{pop,h}$ for specific populations. Define

$$RMSFE_{all} = \sqrt{\frac{1}{I \times X \times (T - T_{jy})} \sum_{t=T_{jy}+1}^{T} \sum_{x=1}^{X} \sum_{i=1}^{I} (y_{x,t,i} - \hat{y}_{x,t,i})^2}, \tag{29}$$

$$RMSFE_{pop_i} = \sqrt{\frac{1}{X \times (T - T_{jy})} \sum_{t=T_{jy}+1}^{T} \sum_{x=1}^{X} (y_{x,t,i} - \hat{y}_{x,t,i})^2}, \tag{30}$$

$$RMSFE_{all,h} = \sqrt{\frac{1}{I \times X} \sum_{x=1}^{X} \sum_{i=1}^{I} (y_{x,h,i} - \hat{y}_{x,h,i})^2}, \tag{31}$$

$$RMSFE_{pop_i,h} = \sqrt{\frac{1}{X} \sum_{x=1}^{X} (y_{x,h,i} - \hat{y}_{x,h,i})^2}, \tag{32}$$

$$MAFE_{all} = \frac{1}{I \times X \times (T - T_{jy})} \sum_{t=T_{jy}+1}^{T} \sum_{x=1}^{X} \sum_{i=1}^{I} |y_{x,t,i} - \hat{y}_{x,t,i}|, \tag{33}$$

where T_{jy} is the jump-off period (associated with 1999 in our empirical analysis, unless specified otherwise), h in (31) and (32) means a specific time step and $T_{jy} + 1 \leq h \leq T$. Here, $y_{x,t,i}$ is the observed value of mortality for age group x, population i, in year t, and $\hat{y}_{x,t,i}$ is the corresponding forecasted value.

Table 4 summarizes the out-of-sample performance. While the multi-population STAR model exhibits good forecasting performance in Group 5, other groups have poor results. This underscores that different populations have distinct spatial–temporal structures, and the STAR model's restrictive and ad hoc structure cannot account for all situations. Our SWVAR model demonstrates lower RMSFE and greater stability than the factor-based Li-Lee model, the more sophisticated Co-FDM model, and other VAR models.

To further assess the performance of the SWVAR model across different horizons, Fig. 3 presents the $RMSFE_{all,h}$. The STAR model performs well in Group 5 but poorly in the other groups. Excluding Group 5, our SWVAR model has the best one-step forecasting and maintains a low $RMSFE_{all,h}$ as the prediction horizon increases. Fig. 4 displays the $RMSFE_{pop,h}$ for each forecasting step h, for some

Table 4
Out-of-sample performance.

Group \ Model / Metric	Li-Lee	Co-FDM	STAR stack	STAR single	VAR	SVAR stack	SVAR single	SWVAR
				Group 1				
RMSFE	0.2533	0.2996	0.2538	0.3158	0.2661	0.2257	0.2249	**0.2166**
σ	0.0377	0.1152	0.0634	0.0654	0.0575	0.0504	0.0515	0.0463
max	0.3324	0.5818	0.3510	0.4370	0.3465	0.3180	0.3195	0.2926
min	0.2055	0.0615	0.1507	0.2171	0.1389	0.1348	0.1370	0.1327
MAFE	0.1956	0.2295	0.1733	0.2275	0.1977	0.1669	0.1647	**0.1520**
				Group 2				
RMSFE	0.3075	0.3416	0.3747	0.2631	0.2457	0.2378	0.2448	**0.2349**
σ	0.1180	0.1687	0.0577	0.0945	0.0675	0.0760	0.0790	0.0703
max	0.4617	0.7238	0.4424	0.3998	0.3016	0.3399	0.3394	0.3135
min	0.1842	0.0776	0.3077	0.1761	0.1338	0.1704	0.1697	0.1640
MAFE	0.2252	0.2567	0.2872	0.1854	0.1731	0.1658	0.1686	**0.1568**
				Group 3				
RMSFE	0.3838	0.3917	0.3191	0.3168	0.2705	0.2905	0.2933	**0.2805**
σ	0.1396	0.1855	0.0678	0.0839	0.0952	0.0769	0.0771	0.0665
max	0.5993	0.9552	0.4248	0.3998	0.4091	0.4064	0.4044	0.3852
min	0.2118	0.0633	0.2415	0.1761	0.1848	0.1843	0.1792	0.2010
MAFE	0.2618	0.2674	0.2100	0.2303	0.1907	0.2008	0.2006	**0.1873**
				Group 5				
RMSFE	0.2588	0.2920	0.2157	0.3129	0.2646	0.2281	0.2269	**0.2134**
σ	0.0406	0.1143	0.0227	0.0795	0.0389	0.0299	0.0303	0.0373
max	0.3232	0.5601	0.2426	0.4370	0.3046	0.2487	0.2506	0.2503
min	0.2126	0.0559	0.1830	0.2171	0.2032	0.1727	0.1713	0.1515
MAFE	0.2052	0.2253	**0.1503**	0.2253	0.2055	0.1742	0.1719	0.1544

Note1: The first row in table is $RMSFE_{all}$, the second row is the standard deviation of $RMSFE_{pop}$. The last two rows are maximum and minimum values of $RMSFE_{pop}$ around all populations. Boldfaced and red values are the highest among all models.

selected populations in Groups 1 and 3. The remaining figures are reported in Online Resource 1 (Figs. S3, S8, S13). This figure shows that the $RMSFE_{pop,h}$ of the SWVAR model is lower than that of the other models in some populations. In some populations and steps, the Li-Lee model and Co-FDM have the lowest $RMSFE_{pop,h}$. However, these two models provide volatile performance. The SWVAR model generally demonstrates its ability to produce accurate forecasts across various horizons.

After that, we show the log-mortality forecasts for the last testing year in Fig. 5 for some countries in Groups 1 and 3. For the other countries in Groups 1 and 3 and Group 2, we report the corresponding figures in Online Resource 1 (Figs. S4, S9, S14). Both the SVAR and SWVAR models get closer to the observed value, and the SWVAR model outperforms in segments where the observed log-mortality rates have large fluctuations.

4.5.2. Robustness analysis

To evaluate the robustness of the forecasting results, we perform an out-of-sample forecasting analysis for two age groups of male and female data within a shorter sample period to compare our proposed SWVAR model with the Li-Lee model, Co-FDM, stacked STAR model, and stacked SVAR model. The different settings are listed in Table 5, and the results are summarized in Table 7 in Appendix E.

Firstly, we conducted an out-of-sample forecast analysis using a shorter training period from 1970 to 1999, testing the robustness of sample coverage. As presented in Table 7, the SWVAR model leads to the most accurate forecasts for Groups 1 to 3 in both matrices. Although the STAR model has a lower RMSFE for Group 5, our SWVAR model still has a lower MAE.

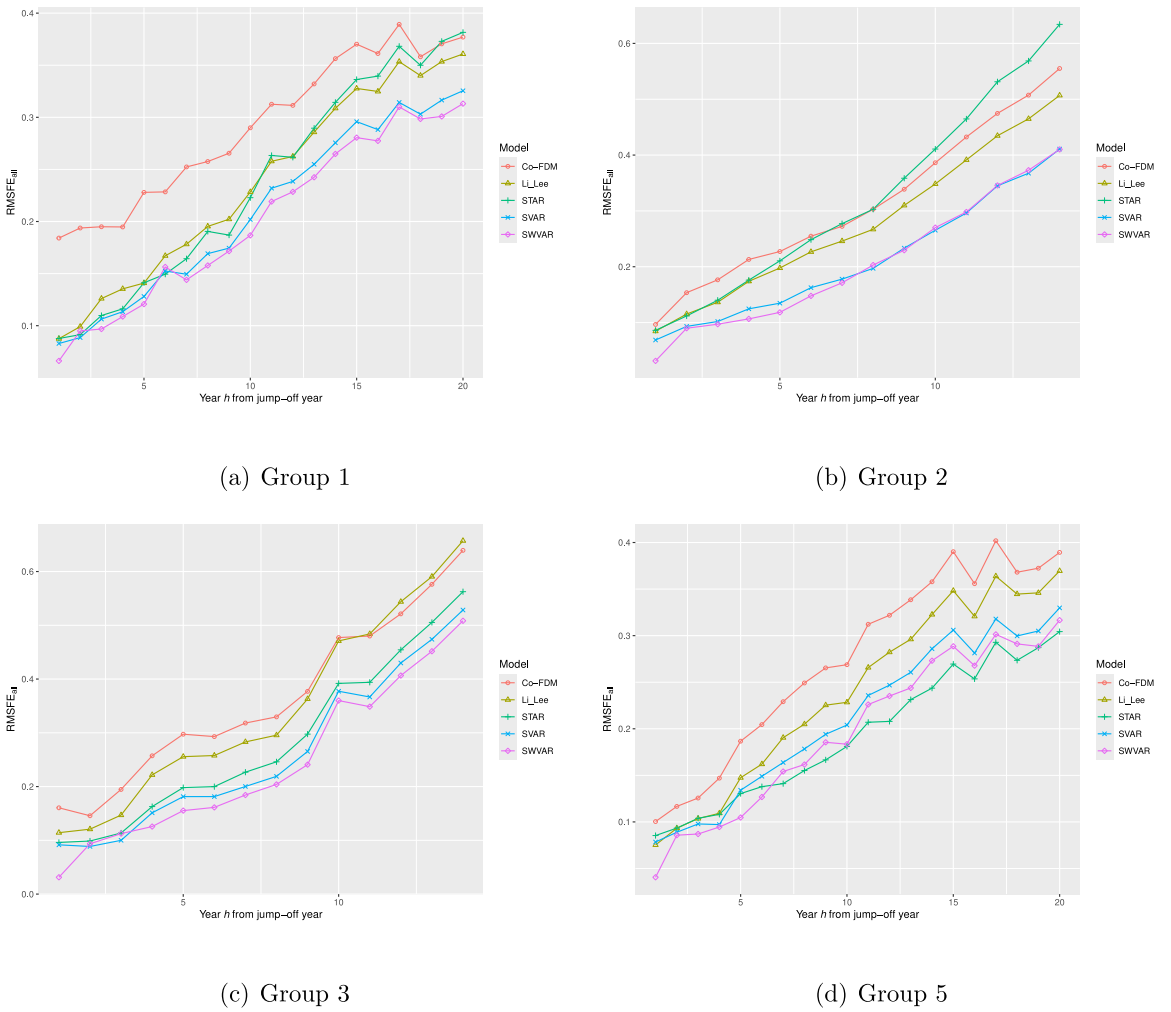
Secondly, we use the single-age data from 41 to 85 to conduct the forecasting process. We select ages 41 to 85 to match the coverage used in Giacometti et al. (2012), Li and Shi (2021) and to test the sensitivity of excluding young and very old ages. We find that our SWVAR model still outperforms in most cases.

Finally, we verify robustness by comparing forecasting results with female and male data. The results for female data differ from those for male data in the one-year age group, in line with the existing finding that cohort effects could differ for male and female mortality (Renshaw & Haberman, 2006). Although not always the best, the SWVAR model has above-average forecasting performance in the male data with a one-year age group. The SWVAR model consistently produces satisfactory forecasting performance across different settings.

4.5.3. Long-run forecasting

We forecast period life expectancies under the uniform distribution of deaths assumption and five-year age cohorts.[6] Period life expectancies are popular because they intuitively summarize the mortality probabilities in one given year. For instance, cohort life expectancies calculate the life expectancy of a particular generation, but this requires forecasting mortality probabilities much further into the future. For each group, we project the period of life expectancies at birth for all populations using the

[6] For more details about the computation of period life expectancy, we refer to Preston et al. (2001) and Wilmoth (2005).



(a) Group 1 (b) Group 2

(c) Group 3 (d) Group 5

Fig. 3. $RMSFE_{all,h}$ of different methods with increasing years h determined from the jump-off year (associated with 1999) for Groups 1, 2, 3, and 5.

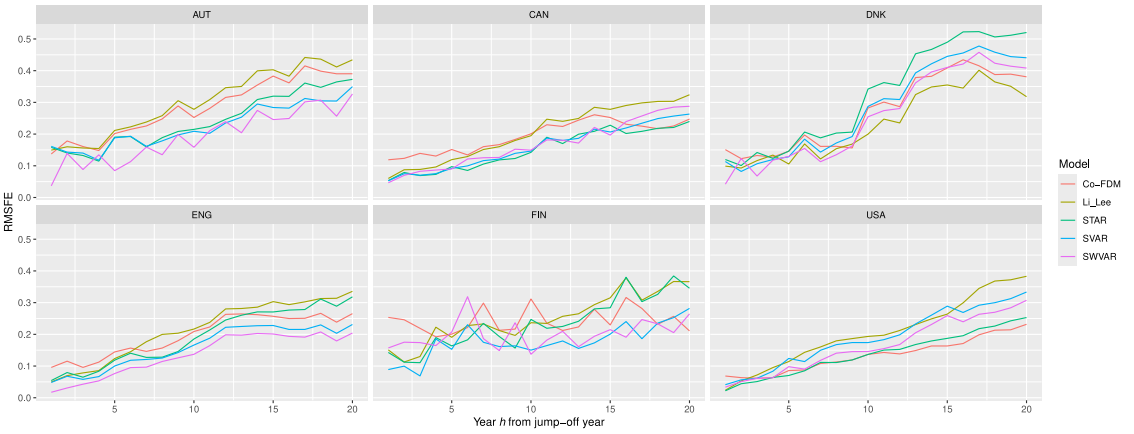
Table 5
Robustness analyses summary.

	Sample Period	Age Group	Gender
Original	1950–1999	Five-year age group (0-99)	Male
Setting 1	1970–1999	Five-year age group (0-99)	Male & Female
Setting 2	1970–1999	One-year age group (41-85)	Male & Female

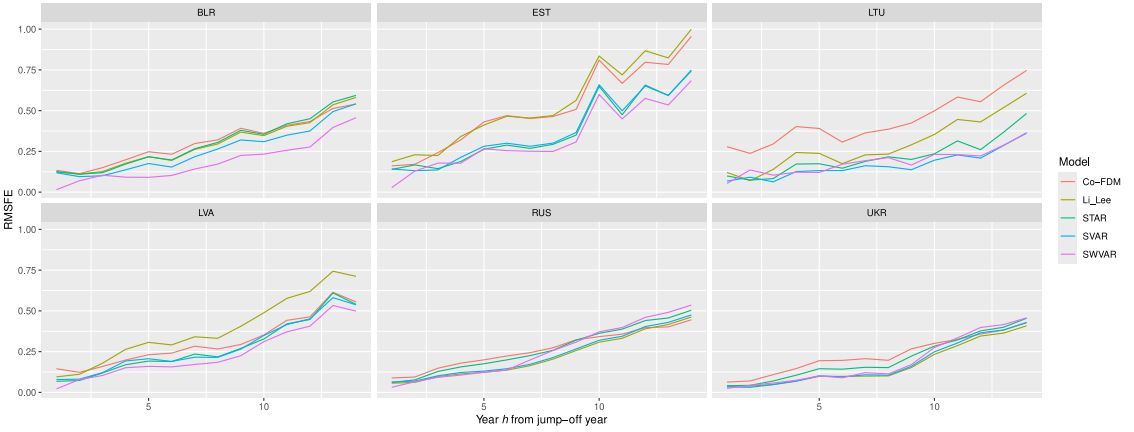
fitted Li-Lee model and the SWVAR model (with training data from 1950–1999 for Group 1 and from 1959–1999 for Groups 2 and 3). The means and 95%-confidence intervals for each population are calculated numerically using 1,000 simulated paths. Fig. 6(a) displays the period life expectancy for several populations in Groups 1 to 3; the remainder is provided in Online Resource 1 (Figs. S5, S10, S15).

In Group 1, most populations exhibit a monotone increasing trend as shown in Fig. 6(a). Consequently, the factor-based Li-Lee and the SWVAR models produce narrow confidence intervals. We cannot evaluate the long-run forecasting performance via the width of the confidence interval because the width greatly varies across populations. For instance, if the SWVAR model learned a large variation from the historical data of the USA, then it produces a wide confidence interval and perfectly matches the observed value. Generally, the SWVAR model's confidence intervals better cover the observed values during the test period compared to the Li-Lee model. Furthermore, the Li-Lee model performs poorly in long-run forecasting for Japan.

In Groups 2 and 3, many populations exhibit volatile historical life expectancies. Due to these complicated historical trends, the Li-Lee and SWVAR models produce wider confidence intervals. Fig. 6(b) shows the superiority of the SWVAR model in such situations. For example, the historical data of Latvia show a long-term decreasing trend with brief increases. Both Li-Lee and SWVAR models



(a) Group 1



(b) Group 3

Fig. 4. $RMSFE_{pop,h}$ of different methods with increasing years h determined from the jump-off year (associated with 1999) for a selection of different populations from different groups.

learned this trend, but the latter generated a higher estimation, although it still seemed far from the test value.[7] This may be attributed to the model's incorporation of trends from other populations. Another observation is the difference between RUS1 and RUS2. In Fig. 6(b), RUS1 presents forecasts from the SWVAR model fitted with Group 2 data, while RUS2 displays forecasts from the SWVAR model fitted with Group 3 data. There are apparent differences between the two groups' predictions for Russia, suggesting that the final results of the model will be largely affected by the selection of populations. Therefore, identifying which populations are most suitable for grouping in multi-population mortality modeling and determining whether including more populations improves the model are important areas for further research.

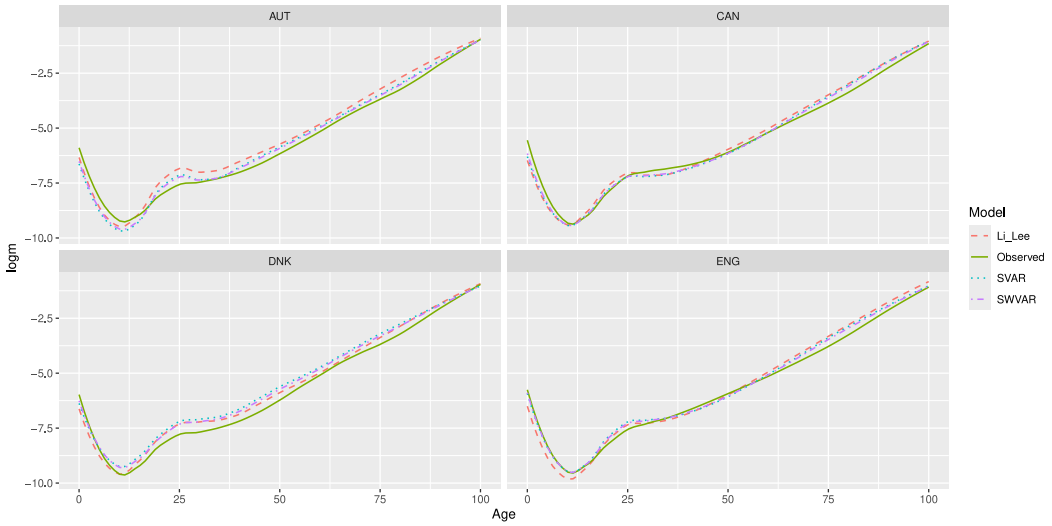
4.5.4. Multi-jump-off years and forecasting performance

Finally, we perform out-of-sample forecasting for each group with various jump-off years to further explore the SWVAR model's forecast performance. We use the RMSFE of log mortality rates and period life expectancy as evaluation metrics. The RMSFE of period life expectancy, as a function of the jump-off year $T_{jy} \leq T-1$, is given by:

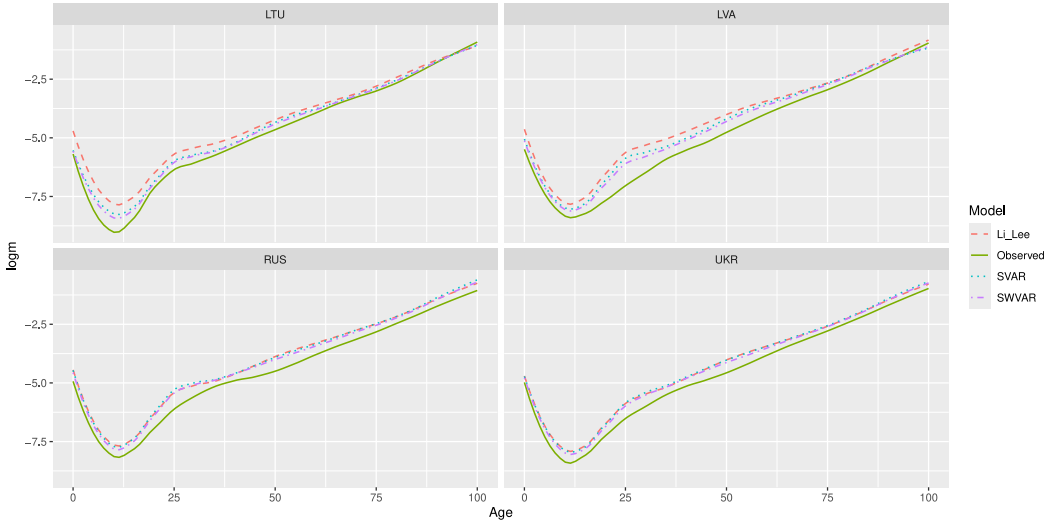
$$RMSFE_{LE}(T_{jy}) = \sqrt{\frac{1}{I \times X \times (T - T_{jy})} \sum_{t=T_{jy}+1}^{T} \sum_{x=1}^{X} \sum_{i=1}^{I} (l_{x,t,i} - \hat{l}_{x,t,i})^2}, \tag{34}$$

where $l_{x,t,i}$ is the period life expectancy at birth for population i in year t, calculated using observed mortality data, and $\hat{l}_{x,t,i}$ is the corresponding forecasted value derived from the model fitted with the data up to the jump-off year (training set).

[7] A more detailed discussion about the forecasts of life expectancy for Latvia and Russia can be found in Appendix F.



(a) Group 1



(b) Group 3

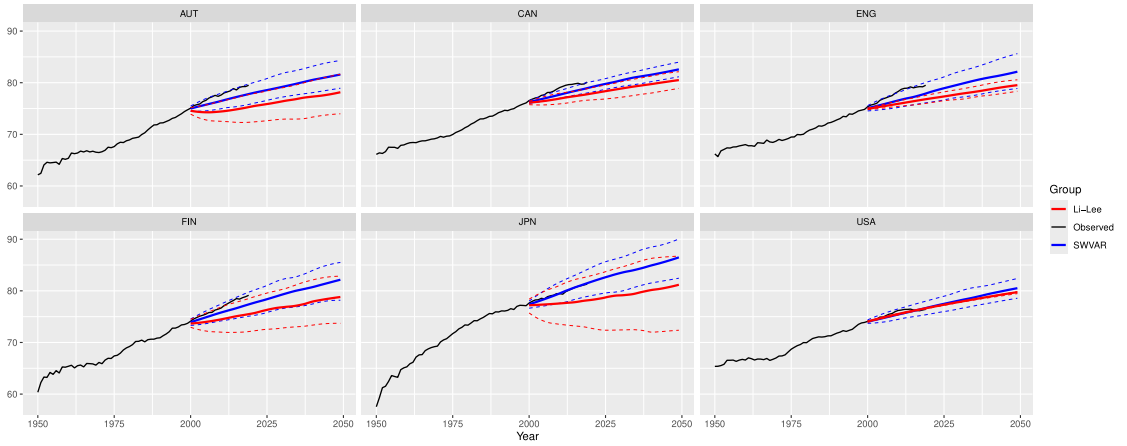
Fig. 5. Forecasts of log mortality for the last test year of different methods. The green solid line labeled "Observed" in the legend is the observed value of the log mortality rate for the last test year. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 7 shows that while the STAR model achieves the best performance in some groups and at some jump-off points, it is extremely unstable and performs poorly as the forecast steps increase. In contrast, the proposed SWVAR model generates better forecasting performance than the Li-Lee, the raw VAR, and the SVAR model, and it exhibits stability in almost all situations.

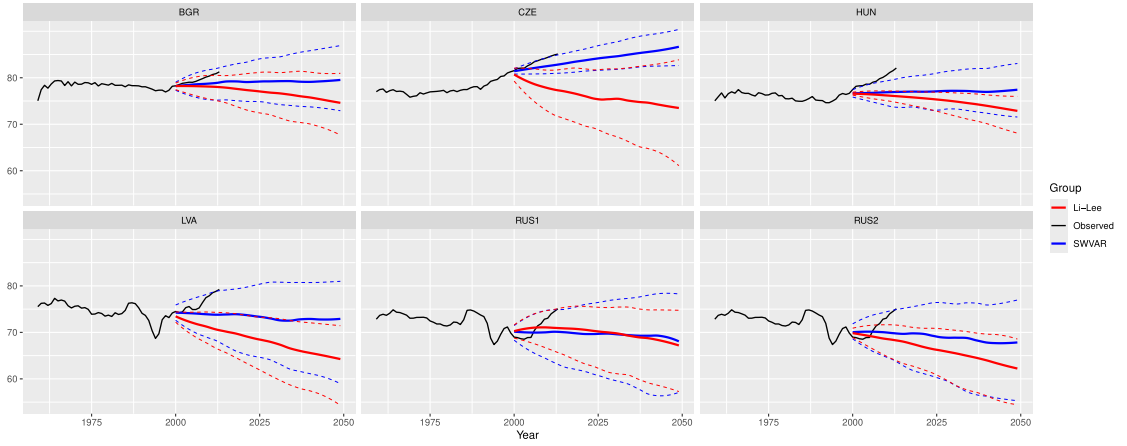
5. Conclusion

This paper introduces the Spatial–temporally Weighted Vector-Autoregressive (SWVAR) model to address multi-population mortality modeling and forecasting problems. To achieve more accurate results, we specifically incorporate an adaptive spatial–temporal weight in the sparse group LASSO estimation.

We demonstrate that classical methods for multi-population mortality modeling often lack adequate calibration. Some models impose excessive constraints, while purely data-driven methods may yield anomalous results despite satisfactory in-sample performance. We then argue that incorporating multiple populations in mortality modeling and spatial–temporal weights enhances the LASSO estimation process, leading to a better model for forecasting. Furthermore, our model is based solely



(a) Group 1



(b) Group 2 and Group 3. In this figure, RUS1 presents forecasts from models fitted with Group 2 data, while RUS2 displays forecasts from models fitted with Group 3 data.

Fig. 6. Period life expectancy. The solid lines are the mean forecasts, and the dashed lines are 95%-confidence intervals. The solid black line labeled "Observed" in the legend is the observed period of life expectancy. We first fit the Li-Lee and SWVAR models with training data (from starting year to 2000), then we use the fitted models to forecast from 2000 to 2050.

on geographical distance and mortality data, which are readily and publicly accessible. As a result, the SWVAR model can model the mortality of multiple populations, offering broad applicability with minimal requirements.

We evaluate the SWVAR model's modeling and forecasting capabilities in the empirical analysis using five multi-population groups. Our findings reveal that spatial–temporal weights significantly improve the fit of a sparse and structured coefficient matrix. Furthermore, the SWVAR model outperforms the Li-Lee, STAR, and SVAR models, as indicated by out-of-sample RMSFE. The SWVAR model provides superior and plausible mortality forecasts while maintaining stability in long-term projections.

We next identify two intriguing avenues for further research. First, although the SWVAR model demonstrates effectiveness using a weighted sparse group LASSO algorithm, its limitations in modeling extremely complex multi-populations highlight the need for a more nuanced exploration of the spatial structure. Second, the method's substantial computational demands for hyperparameter grid searching suggest the need for more efficient optimization techniques.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

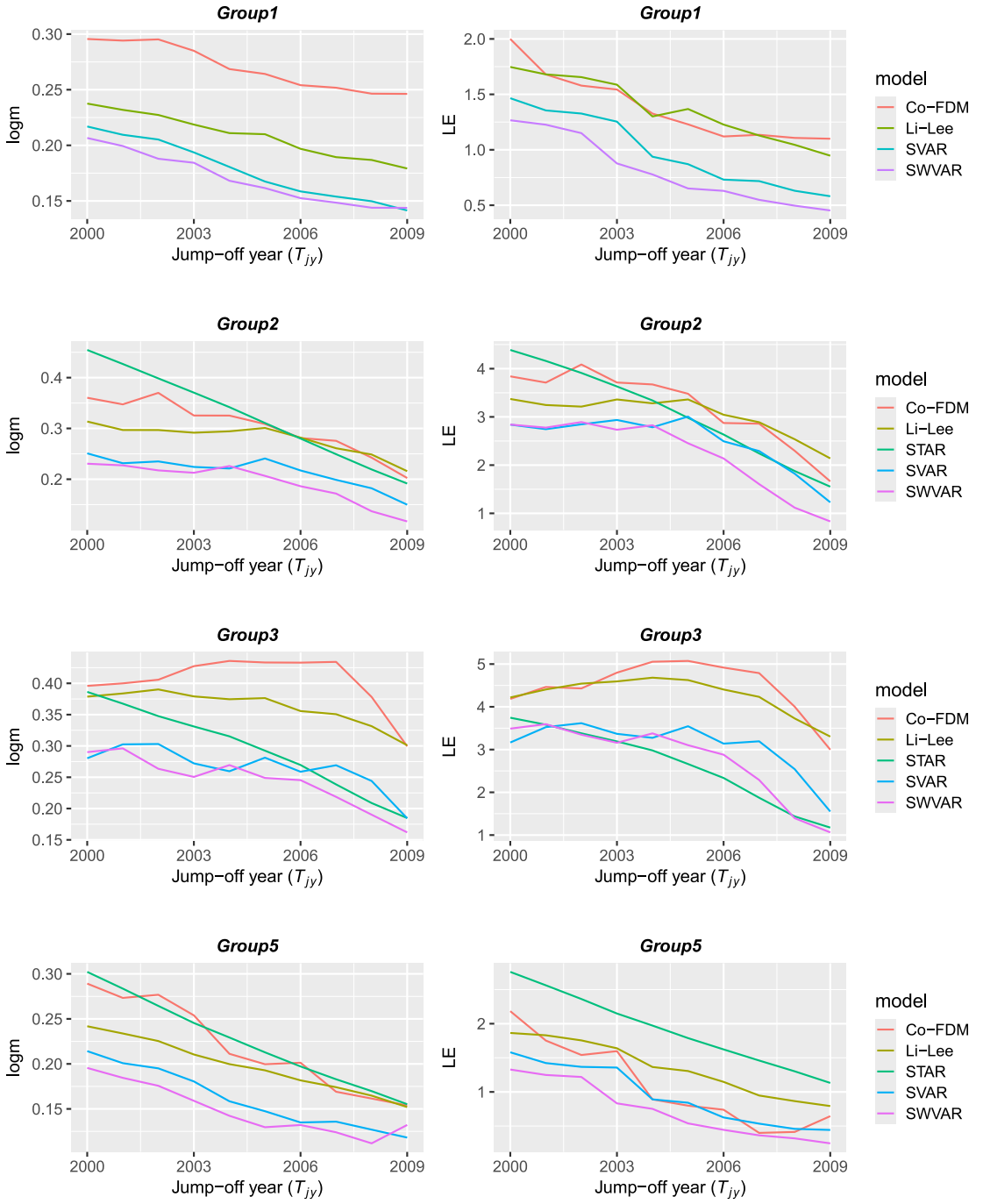
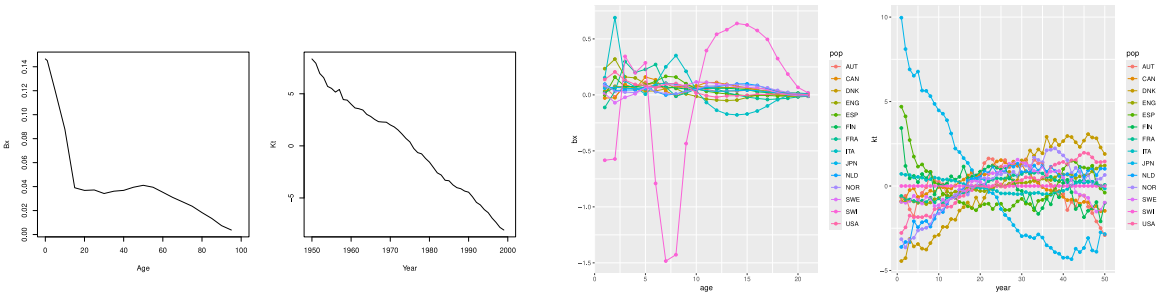


Fig. 7. The relative $RMSFE_{all}$ (left) and $RMSFE_{LE}$ (right) with different jump-off years, as defined in (29) and (34), respectively. We omit the result of the STAR model for Group 1 due to numerical instability.



(a) Common Factor

(b) Population-specific factors

Fig. 8. Estimated parameters of the Li-Lee model in Group 1.

Appendix A. Li-Lee model

Let $y_{x,t,i} = log(m_{x,t,i})$. The Li-Lee model (Li & Lee, 2005), also called the augmented common factor (ACF) model, extends the Lee–Carter model (Lee & Carter, 1992) to multi-population scenarios. They use common factor $B_x K_t$ to describe the common trend in mortality change, population-specific factor $b_{x,i}\kappa_{t,i}$ to describe the difference of mortality between population i and the common trend. Specifically, the ACF model is defined as follows:

$$y_{x,t,i} = a_{x,i} + B_x K_t + b_{x,i}\kappa_{t,i} + \epsilon_{x,t,i}, \quad \epsilon_{x,t,i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2_{\epsilon,x,i}), \tag{35}$$

for age groups $x = 1, \ldots, X$, time periods $t = 0, 1, \ldots, T$, populations $i = 1, 2, \ldots, I$, and $\mathcal{N}(\cdot)$ denotes the Gaussian distribution. To guarantee that in the long run, mortality forecasts converge to the common trend K_t, $\kappa_{i,t}$ needs to be stationary, and K_t contains the trend in future mortality dynamics. Therefore, for forecasting, Lee and Carter (1992) fit the K_t with a random walk with drift (RWD) model and fit the $\kappa_{i,t}$ with AR(1) models:

$$\begin{aligned} K_t &= c + K_{t-1} + \eta_t, \quad \eta_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2_\eta), \\ \kappa_{t,i} &= c_{i,0} + c_{i,1}\kappa_{t-1,i} + \omega_{t,i}, \quad \omega_{t,i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2_{\omega,i}). \end{aligned} \tag{36}$$

The following normalization constraints are imposed:

$$\sum_{x=1}^{X} B_x = \sum_{x=1}^{X} b_{x,i} = 1, \text{ and } \sum_{t=1}^{T} K_t = \sum_{t=1}^{T} \kappa_{t,i} = 0,$$

for all $i = 1, 2, \ldots, I$.

As an illustration, the parameter estimates of the Li-Lee model for populations in Group 1 are shown in Fig. 8, where Group 1 is defined in Appendix D.

Appendix B. Coherent functional demographic model

In this appendix, we define the Co-FDM model (Hyndman et al., 2013). In the Hyndman–Ullah (HU) model (Hyndman & Ullah, 2007), the log-mortality rate is assumed to be a smooth age function observed with error. That is,

$$y_{x,t} = f_t(x) + \sigma_t(x)\epsilon_{x,t}, \tag{37}$$

where $y_{x,t}$ is the observed log-mortality rate at age x and year t, and $f_t(x)$ is an underlying continuous and smooth function, estimated via penalized regression splines with a partial monotonic constraint. Moreover, $\sigma_t(x)$ further allows noise to vary with x, whereas $\epsilon_{x,t}$ is an independently and identically distributed normal random variable.

We next decompose the fitted $f_t(x)$ via a basis function expansion using the following model:

$$f_t(x) = \mu(x) + \sum_{k=1}^{K} \beta_{t,k}\rho_k(x) + e_t(x), \tag{38}$$

where $\mu(x)$ is a time-invariant measure of location of $f_t(x)$ and $\phi_k(x)$ is a set of orthonormal basis functions and $e_t(x) \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, v(x))$.

In Co-FDM, we still assume that there is an underlying smooth function $f_{t,i}(x)$ that we are observing with error. Thus,

$$y_{x,t,i} = log[f_{t,i}(x)] + \sigma_{t,i}(x)\epsilon_{x,t,i}, \tag{39}$$

where x is the age group, $\epsilon_{x,t,i}$ is an independent and identically distributed standard normal random variable, and $\sigma_{t,i}(x)$ allows the amount of noise to vary with age x.

Rather than modeling the original population-specific death rates, the Co-FDM method models and forecasts the geometric mean of all I subpopulation rates and the ratio of each subpopulation rate to product rates. Let

$$\begin{aligned} p_t(x) &= [f_{t,1}(x), f_{t,2}(x), \ldots, f_{t,I}(x)]^{1/I}, \\ r_{t,i}(x) &= f_{t,j}(x)/p_t(x), \end{aligned} \tag{40}$$

where $i = 1, 2, \ldots, I$, $f_{t,i}(x)$ is the underlying continuous and smooth function corresponding to population i and $p_t(x)$ is the geometric mean of the smoothed rates and so represents the joint (non-stationary) behavior of all subpopulations.

Then, the Co-FDM model uses functional time series models for $p_t(x)$ and each $r_{t,i}(x)$ for $i = 1, 2, \ldots, I$:

$$\begin{aligned} log[p_t(x)] &= \mu_p(x) + \sum_{k=1}^{K} \beta_{t,k}\rho_k(x) + e_t(x), \\ log[r_{t,i}(x)] &= \mu_{r,i}(x) + \sum_{l=1}^{L} \gamma_{t,l,i}\psi_{l,i}(x) + w_{t,i}(x), \end{aligned} \tag{41}$$

where $\mu_p(x)$ is a measure of location of $log[p_t(x)]$ and $\mu_{r,i}(x)$ is a measure of location of $log[r_{t,i}(x)]$. $\rho_k(x)$ and $\psi_{l,i}(x)$ are sets of orthonormal basis functions. Here, $e_t(x)$ and $w_{t,i}(x)$ are error terms with zero mean and serially uncorrelated. Consequently, the implied model for each group is given by:

$$log[f_{t,i}(x)] = log[p_t(x)r_{t,i}(x)] = \mu_i + \sum_{k=1}^{K} \beta_{t,k}\phi_k(x) + \sum_{l=1}^{L} \gamma_{t,l,i}\psi_{l,i}(x) + z_{t,i}(x), \quad (42)$$

where $\mu_i(x) = \mu_p(x) + \mu_{r,i}(x)$ is the group mean and $z_{t,i}(x) = e_t(x) + w_{t,i}(x)$ is the error term.

Next, univariate time series models, ARIMA models for β_k and ARMA or ARFIMA models for $\gamma_{l,i}$, $i = 1, 2, \ldots, I$, are implemented to produce the h-steps-ahead forecasts. (of $\beta_{T+h,k}$ and $\gamma_{T+h,l,i}$.) Use the forecast coefficients with (42) to obtain forecastes of $log[f_{t,i}(x)]$, $t = T+1, T+2, \ldots, T+h$. From (39), we subsequently obtain forecasts of $y_{x,t,i}$.

In Hyndman et al. (2013), β_k are forecasted using possibly non-stationary autoregressive integrated moving average (ARIMA)(p, d, q) model (Shumway & Stoffer, 2006) and $\gamma_{l,i}$ are each forecasted using any stationary autoregressive moving average (ARMA)(p, q) model (Box et al., 2008) or autoregressive fractionally integrated moving-average (ARFIMA)(p, d, q) model (Granger & Joyeux, 1980). In our empirical results, we follow the default settings of the **vital** package, that is $K = L = 6$, and we search through the parameter space to identify the optimal ARIMA models for β_k and each $\gamma_{l,i}$, $i = 1, 2, \ldots, I$, by minimizing the small-sample corrected Akaike Information Criterion (AICc).

Appendix C. Model selection tests

Considering initially the true model $f(\cdot)$ of which we wish to test:

$$\mathbf{H0}: y_i = f_i(X_i, \beta) + \epsilon_{0i}, \quad \epsilon_{0i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2_{\epsilon_0}). \quad (43)$$

Suppose that some theory suggests an alternative model $g(\cdot)$:

$$\mathbf{H1}: y_i = g_i(Z_i, \gamma) + \epsilon_{1i}, \quad \epsilon_{1i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2_{\epsilon_1}). \quad (44)$$

The C-test is based on the regression with constraint:

$$y_i = \beta\hat{f}_i + \alpha\hat{g}_i + \epsilon_i, \quad \epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2_{\epsilon_{C1}}), \quad s.t. \quad \alpha + \beta = 1, \quad (45)$$

where $\hat{f}_i = f_i(X_i, \hat{\beta})$ and $\hat{\beta}$ is the estimation of β, $\hat{g}_i = g_i(Z_i, \hat{\gamma})$ and $\hat{\gamma}$ is the estimation of γ. We can turn constrained regression (45) to an unconstrained regression (46) with transformation:

$$y_i - \hat{f}_i = \alpha(\hat{g}_i - \hat{f}_i) + \epsilon_i, \quad \epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2_{\epsilon_{C1}}). \quad (46)$$

If the $\hat{\alpha}$ is negative, this means that $f(\cdot)$ is the correct model; we should therefore directly reject **H1**. If positive, we can perform a t-test to α in the standard regression model (46) to test the effect of $g(\cdot)$. For our problem, let the SVAR model be $f(\cdot)$ and the SWVAR model be $g(\cdot)$; we can apply the t-test to α to evaluate the effectiveness of the SWVAR model.

C-test

1: **H0**: $\alpha = 0$, **H1**: $\alpha \neq 0$.
2: $t_\alpha = \frac{\hat{\alpha}}{\sqrt{var(\hat{\alpha})}}$.
3: Under **H0**, $t_\alpha \sim t(n-1)$.
4: If the p-value is smaller than the significance level, reject the null hypothesis: $\alpha \neq 0$.

However, our problem's characteristics lead to the t-test's imprecision. The standard errors are not particularly meaningful for strongly biased estimates since penalized estimation is a procedure that reduces the variance of estimators by introducing significant bias. Therefore, the t-test of a penalized estimate provides only a partial picture.

As a refinement of the C-test, the J-test is based on the regression:

$$y_i = (1-\alpha)f_i(X_i, \beta) + \alpha\hat{g}_i + \epsilon_i, \quad \epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2_{\epsilon_j}), \quad (47)$$

where $\hat{g}_i = g_i(Z_i, \hat{\gamma})$ and $\hat{\gamma}$ is the estimation of γ. If **H0** in (43) is true, the real value of α will be 0. Furthermore, the regressive Eq. (47) for J-test can be transformed to a VARX(1) model:

$$\mathbf{y}_t = A\mathbf{y}_{t-1} + \alpha\mathbf{w}_{t-1} + \mathbf{u}_t, \quad \mathbf{u}_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \Sigma^2_u). \quad (48)$$

In Hamilton (2020), a goodness-of-fit test was introduced to evaluate the effect of the exogenous variable, effectively substituting the t-test to avoid using variances. By taking the estimation of the SWVAR model as the exogenous variable $\mathbf{w}_t$, we can evaluate the effect of the SWVAR model under the null hypothesis that the VAR model is effective.

J-test

1: **H0**: $\alpha = 0$, **H1**: $\alpha \neq 0$.
2: $\Lambda = 2[\log L(\hat{\theta}) - \log L(\hat{\theta}_{\alpha=0})]$.
3: Under **H0**, $\Lambda \sim \chi^2(1)$.
4: If the p-value is smaller than the significance level, reject the null hypothesis: $\alpha \neq 0$.

Appendix D. Population list

We list the countries in all groups that we select in this paper in Table 6.

Appendix E. Robustness analysis

We present the summarized results of robustness analysis in Table 7.

Appendix F. Life expectancy in Latvia and Russia

Fig. 6(b) suggests that the life expectancy forecasts are inaccurate for Latvia (LVA) and Russia (RUS, in Groups 2

Table 6
Complete list of all populations in each group.

Group number	Populations			
1	Austria (AUT) Finland (FIN) The Netherlands(NLD) Switzerland (SWI)	Canada (CAN) France (FRA) Norway (NOR) United States of America (USA)	Denmark (DNK) Italy (ITA) Spain (ESP)	England (ENG) Japan (JPN) Sweden (SWE)
2	Bulgaria (BGR) Lithuania (LTU)	Czech Republic (CZE) Russia (RUS)		Hungary (HUN)
3	Belarus (BLR) Russia (RUS)	Estonia (EST) Ukraine (UKR)	Latvia (LVA)	Lithuania (LTU)
4	Belarus (BLR) Estonia (EST) Russia (RUS)	Bulgaria (BGR) Hungary (HUN) Ukraine (UKR)	Czech Republic (CZE) Latvia (LVA)	Lithuania (LTU)
5	Austria (AUT) Japan (JPN)	England (ENG) The Netherlands(NLD)	France (FRA) Switzerland (SWI)	Italy (ITA)

Note: We list the countries in all groups that we select in this paper. The populations show the abbreviations; we use those in our figures for brevity.

Table 7
Robustness analysis.

Setting / Model / Metric	Li-Lee	Co-FDM	STAR	SVAR	SWVAR	Li-Lee	Co-FDM	STAR	SVAR	SWVAR
	Female					Male				
	Group 1, Setting 1									
RMSFE	**0.1844**	0.2320	0.3665	0.1953	0.2005	0.2231	0.2622	0.2285	0.2230	**0.2135**
MAFE	**0.1317**	0.1696	0.2526	0.1343	**0.1314**	0.1648	0.2007	0.1569	0.1563	**0.1436**
RMSFE	0.2674	0.2534	0.3075	0.2195	**0.1986**	0.3154	0.3091	0.3162	0.2514	**0.2464**
MAFE	0.2028	0.1949	0.2380	0.1547	**0.1351**	0.2341	0.2265	0.2209	0.1725	**0.1625**
RMSFE	0.3766	0.3857	0.3413	0.3080	**0.2692**	0.4025	0.3638	0.3102	0.3104	**0.2939**
MAFE	0.2610	0.2754	0.2306	0.2123	**0.1810**	0.2672	0.2497	0.2039	0.2146	**0.1954**
RMSFE	0.1773	0.1978	0.4592	0.1807	**0.1712**	0.2348	0.2254	**0.1923**	0.2141	0.1982
MAFE	0.1275	0.1436	0.317	0.1248	**0.1127**	0.1766	0.1715	0.1389	0.1546	**0.1362**
	Group 1, Setting 2									
RMSFE	0.1791	0.1949	0.2920	0.1672	**0.1629**	0.1967	0.1705	0.2758	0.1770	**0.1623**
MAFE	0.1432	0.1460	0.1899	0.1170	**0.1143**	0.1595	0.1312	0.1934	0.1340	**0.1166**
RMSFE	0.2524	0.1947	0.2601	0.1846	**0.1838**	0.2685	0.2126	**0.1481**	0.2366	0.2427
MAFE	0.1984	0.1527	0.1770	0.1386	**0.1325**	0.1902	0.1587	**0.1038**	0.1664	0.1661
RMSFE	0.2637	0.2461	0.2629	**0.2342**	0.2439	0.2687	**0.2124**	0.2684	0.2815	0.2647
MAFE	0.1945	0.1851	0.1937	**0.1745**	0.1761	0.1878	**0.1572**	0.1965	0.1986	0.1780
RMSFE	0.1255	**0.1161**	0.1251	0.1368	0.1347	0.1598	**0.1239**	0.1460	0.1646	0.1485
MAFE	0.0954	**0.0897**	0.0936	0.0989	0.0952	0.1313	**0.1005**	0.1160	0.1291	0.1100

Note 1: This table displays the overall RMSFE and MAFE across all ages, populations, and time horizons. Boldfaced and red values are the highest among all models.
Note 2: The result of the STAR model for Group 1, Setting 2 is not optimal because the computer is running out of memory when solving the constrained optimization problem.

and 3) for both the SWVAR and Li-Lee models. Therefore, we test several well-known alternative approaches in the literature, including the Lee–Carter model, the HU model, the Co-FDM model, the Li-Lee model, the STAR model, and the SVAR model.

As shown in Fig. 9, none of these methods perform ideally with the Russian and Latvian datasets. Furthermore, we observe substantial gaps between the one-step forecasts of different approaches and the corresponding observed value. We attribute these observations to the two populations' life expectancy fluctuation in the sample period. Meanwhile, since these two life expectancies present general decreasing trends in the sample period, any VAR models with lag order one will naturally generate a decreasing forecast in the long run. These findings suggest that the challenges we face are not unique to our method but are common across several established approaches.

Given these results, we believe that forecasting life expectancy for populations with heavily fluctuating mortality dynamics remains a complex problem, and no current method provides an entirely satisfactory solution in this domain. We leave this as a topic for further research.

Appendix G. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.03.004.

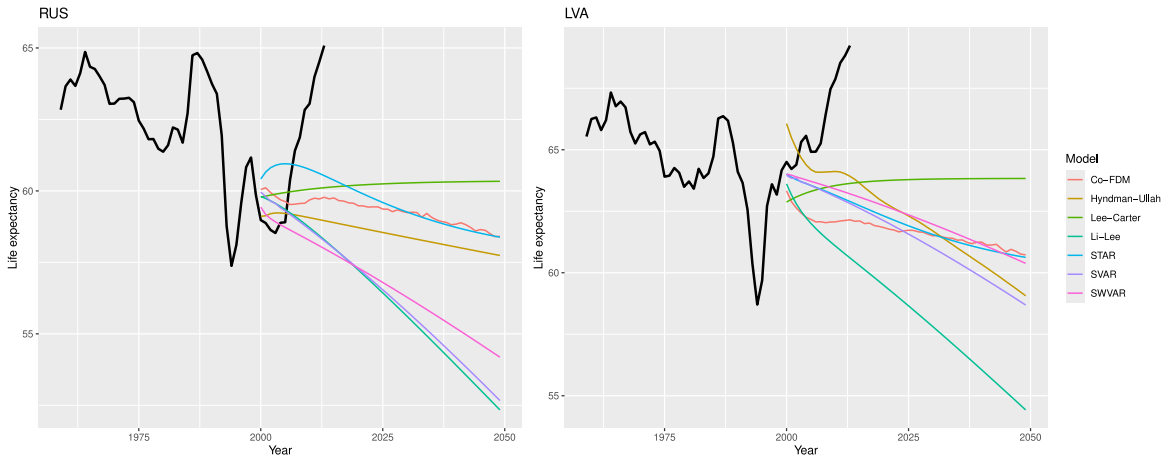


Fig. 9. Period life expectancy of RUS (left) and LVA (right). All multi-population models are fitted for Group 3. The black lines are life expectancies computed from observed values, and different colored lines forecast different models' values. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Data and code availability

Data and replication package for this paper are available at https://github.com/Greatknee/Replication-SWVAR.

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